

HiSIM_SOTB 1.3.0 User's Manual

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HiSIM_SOTB Developers

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1 Definition of the Device Parameters

HiSIM_SOTB is a descendant of HiSIM_SOI valid for ultra-thin SOI and BOX layers applicable even for the double gate structure as a specific structure of SOTB-MOSFET.

HiSIM_SOI itself is already valid for any possible SOTB-MOSFET structures, including different types of charges induced within the device. In addition to the HiSIM_SOI treatment, HiSIM_SOTB considers charges induced at both sides of BOX explicitly within the Poisson equation.

As depicted in Fig. 1, the definition of the node potentials are:

- V_s : reference
- V_{gs} : gate to source
- V_{ds} : drain to source
- V_{es} : substrate to source

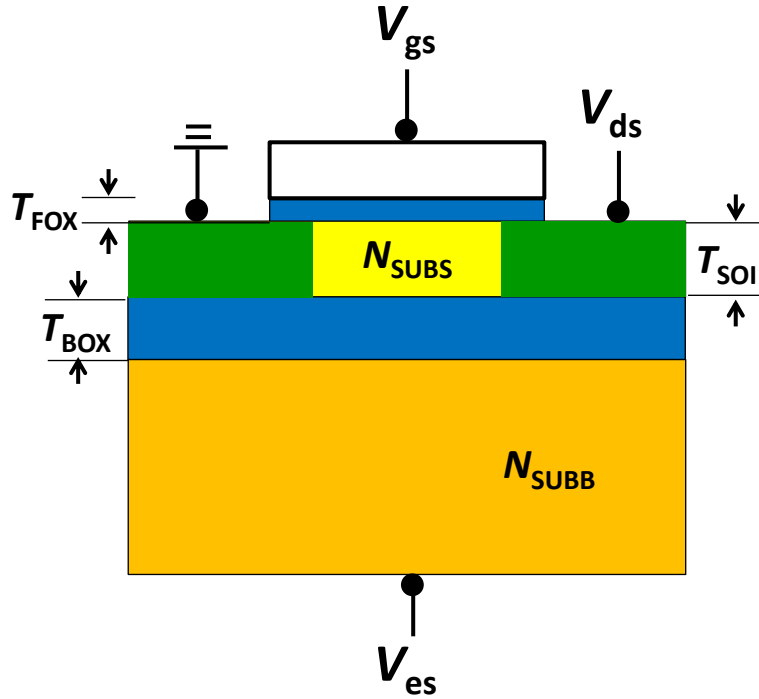


Fig. 1: Definitions of nodes and device parameters in HiSIM_SOTB. The y-axis is defined along the direction of current flow (source is $y=0$) and x is defined downwards into the channel (front gate MOS interface is $x=0$).

The effective channel length L_{eff} and width W_{eff} are calculated from the gate length L_{gate} and width W_{gate} , where L_{gate} and W_{gate} are related to the drawn gate length L_{drawn} and width W_{drawn} as

$$L_{gate} = L_{drawn} \quad (1)$$

$$W_{gate} = \frac{W_{drawn}}{NF} \quad (2)$$

$$L_{\text{eff}} = L_{\text{gate}} - 2 \times X_{\text{ld}} \quad (3)$$

$$W_{\text{eff}} = W_{\text{gate}} - 2 \times \mathbf{XWD} - 2 \times W_{\text{STI,mod}} \quad (4)$$

$$W_{\text{effc}} = W_{\text{gate}} - 2 \times \mathbf{XWDC} - 2 \times W_{\text{STI,mod}} \quad (5)$$

$$X_{\text{ldmod}} = \frac{\mathbf{XLD} \cdot L_{\text{gate}}}{\mathbf{XLDL} + \mathbf{XLDLMIN}} \quad (6)$$

$$X_{\text{ld}} = \begin{cases} \min(X_{\text{ldmod}}, \mathbf{XLD}) & (\mathbf{XLD} > 0) \\ \max(X_{\text{ldmod}}, \mathbf{XLD}) & (\mathbf{XLD} \leq 0) \end{cases} \quad (7)$$

where **XLD** and **XWD** account for the overlaps of source/drain contact and the gate oxide. The model parameter **XWDC** is introduced to describe the different width dependence of capacitance from currents. **NF** is the number of fingers.

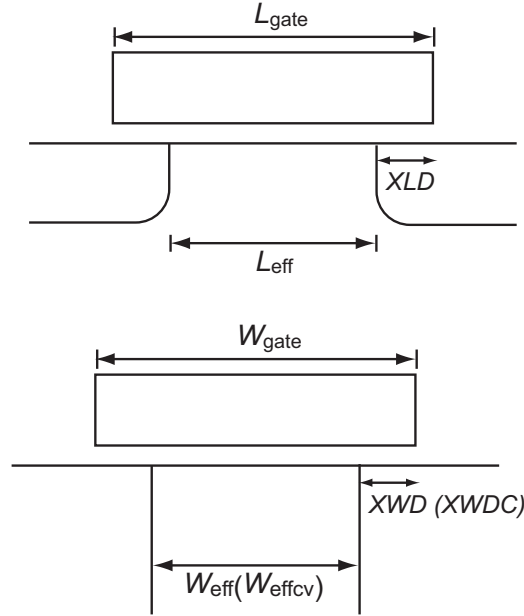


Fig. 2: Cross section of the device.

Table 1: HiSIM_SOTB model parameters introduced in Section 1 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

NF	number of gate fingers
XLD	gate overlap length
XLDL	channel-length dependence of XLD
XLDLMIN	minimum value of XLDL
XWD	gate overlap width
XWDC	different width dependence of capacitance from currents

2 Basic Equations Solved in the HiSIM_SOTB Core Part

Though all following descriptions are written for n-channel MOSFET, they are also valid for the p-channel case. The Poisson equation solved in HiSIM_SOTB together with the Gauss law is written:

$$\phi_s = V'_G + \frac{Q_{\text{bulk}} + Q_i + Q_{\text{dep,SOI}} + Q_b + Q_h}{C_{\text{FOX}}} \quad (8)$$

$$V'_G = V_{\text{gs}} - V_{\text{FB}} + \Delta V_{\text{th}} \quad (9)$$

where ϕ_s is the surface potential at the front gate and V_{FB} is the flat-band voltage with the threshold voltage shift ΔV_{th} in comparison to the threshold voltage of a long-channel transistor [4] (see Section 4). Charges Q 's induced within the SOTB-MOSFET are depicted in Fig. 3 and Q_h is the excess charge in the SOI layer due to the impact ionization (see Section 17). Additional equations characterizing the potential distribution to the device depth direction are derived with boundary conditions at different surfaces.

Following three different conditions for describing the potential distribution are considered.

(i) $Q_{\text{bulk}} < 0$: smooth potential change along the device depth direction

$$\begin{aligned} \phi_s &= \phi'_b + \frac{Q_{\text{bulk}} + \frac{1}{2}Q_{\text{dep,SOI}} + Q_b}{C_{\text{SOI}}} \\ \phi_b &= \phi'_b \end{aligned} \quad (10)$$

(ii) $Q_{\text{bulk}} > 0$: smooth potential change along the device depth direction

$$\begin{aligned} \phi_b &= \phi'_b + \frac{Q_{\text{bulk}} - \frac{1}{2}Q_{\text{dep,SOI}}}{C_{\text{SOI}}} \\ \phi_s &= \phi'_b \end{aligned} \quad (11)$$

(iii) $Q_{\text{bulk}} \approx 0$: potential peak induced within the SOI layer (transition region between i and ii)

$$\begin{aligned} \phi_s &= \phi'_b + \frac{Q_{\text{bulk}} + \frac{1}{2}Q_{\text{dep,SOI}} + Q_b}{C_{\text{SOI}}} \\ \phi_b &= \phi'_b + \frac{Q_{\text{bulk}} - \frac{1}{2}Q_{\text{dep,SOI}}}{C_{\text{SOI}}} \end{aligned} \quad (12)$$

An additional equation is derived with the Gauss law at BOX as

$$\phi_{\text{bulk}} = \phi_b + \frac{Q_{\text{bulk}}}{C_{\text{BOX}}} \quad (13)$$

Definitions for various capacitances are

$$C_{\text{FOX}} = \frac{\epsilon_{\text{ox}}}{\text{TFOX}} \quad (14)$$

$$C_{\text{BOX}} = \frac{\epsilon_{\text{ox}}}{\text{TBOX}} \quad (15)$$

$$C_{\text{SOI}} = \frac{\epsilon_{\text{Si}}}{\text{TSOI}} \quad (16)$$

where **TFOX** is the physical gate-oxide thickness and $\epsilon_{\text{ox}} (= \epsilon_0 \cdot \kappa)$ is the permittivity of dielectrics. ϵ_0 and κ are the permittivity of vacuum and the dielectric constant, respectively. The BOX thickness and the SOI layer thickness are denoted by **TBOX** and **TSOI**. The expression of charges will be shown readily on next page.

Eqs. (8)– (13) are solved simultaneously within HiSIM_SOTB with the Newton iteration method. In this solution, ϕ_s , ϕ_b , and ϕ_{bulk} are obtained. The solution takes place at source side and drain side. Prior to the Newton iteration, an approximated, analytical initial value for each variable is estimated in the code.

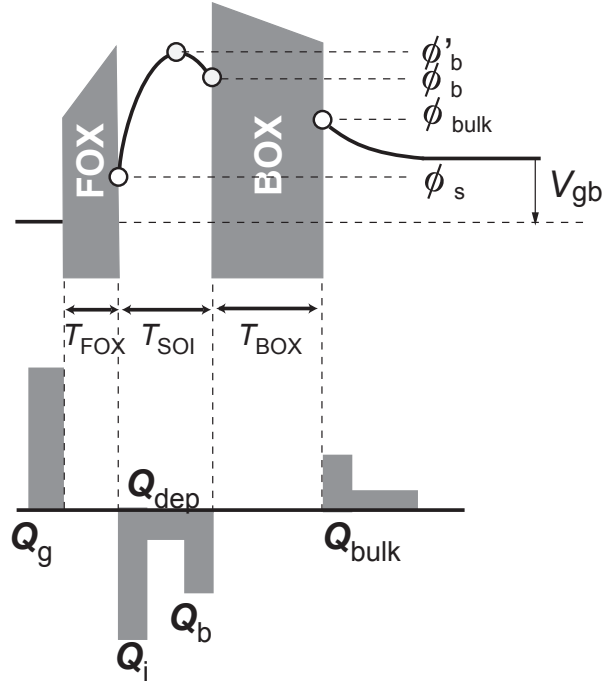


Fig. 3: Band structure of SOTB-MOSFET and generated charges.

All device characteristics are determined on the basis of charges controlled by applied voltages. The charges induced are functions of surface potentials written as

$$Q_{\text{dep,SOI}} = -qN_{\text{subs}} \cdot \text{TSOI} = Q_{\text{s,dep}} + Q_{\text{b,dep}} \quad (17)$$

$$Q_{\text{s,dep}} = -\sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subs}}}{\beta}} \left[\exp(-\beta\phi_{\text{s}}(y)) + \beta \{ \phi_{\text{s}}(y) - \phi'_{\text{b}}(y) \} - 1 \right]^{\frac{1}{2}} \quad (18)$$

$$Q_{\text{b,dep}} = -\sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subs}}}{\beta}} \left[\exp(-\beta\phi_{\text{b}}(y)) + \beta \{ \phi_{\text{b}}(y) - \phi'_{\text{b}}(y) \} - 1 \right]^{\frac{1}{2}} \quad (19)$$

$$Q_{\text{i}} = -\sqrt{Q_{\text{s,dep}}^2 + \frac{n_{\text{p0}}}{p_{\text{p0}}} \{ \exp(\beta(\phi_{\text{s}} - \phi_{\text{f}})) - \exp(\beta(\phi'_{\text{b}} - \phi_{\text{f}})) \}} - Q_{\text{s,dep}} \quad (20)$$

$$Q_{\text{b}} = -\sqrt{Q_{\text{b,dep}}^2 + \frac{n_{\text{p0}}}{p_{\text{p0}}} \{ \exp(\beta(\phi_{\text{b}} - \phi_{\text{f}})) - \exp(\beta(\phi'_{\text{b}} - \phi_{\text{f}})) \}} - Q_{\text{b,dep}} \quad (21)$$

$$Q_{\text{bulk}} = -\sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subb}}}{\beta}} \left[\exp\{-\beta(\phi_{\text{bulk}}(y) - V_{\text{es}})\} + \beta(\phi_{\text{bulk}}(y) - V_{\text{es}}) - 1 \right. \\ \left. + \frac{n_{\text{p0}}}{p_{\text{p0}}} \{ \exp(\beta\phi_{\text{bulk}}(y)) - \exp(\beta V_{\text{es}}) \} \right]^{\frac{1}{2}} \quad (22)$$

where ϕ_{f} is quasi-Fermi potential at the location y , N_{subs} and N_{subb} are modified impurity concentration of SOI layer and substrate (see Eqs 65 and 66 in Section 4), respectively. Please be reminded that the y -axis is defined along the direction of current flow (source is $y=0$), as was in Fig. 1.

For later reference in this manual, the factors in the above equations will be represented as:

$$\text{const0} = \sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subs}}}{\beta}} \quad (23)$$

$$\text{const0bulk} = \sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subb}}}{\beta}} \quad (24)$$

In equations 18 through 24,

$$\beta = \frac{q}{kT} \quad (25)$$

where q is the electron charge. The Boltzmann constant and the lattice temperature in Kelvin are k and T , respectively. The electron concentration at equilibrium condition n_{p0} is

$$n_{p0} = \frac{n_i^2}{p_{p0}} \quad (26)$$

where n_i is the intrinsic carrier concentration and p_{p0} is the equilibrium concentration of the majority carrier in the substrate, hole for n-MOSFETs and electrons for p-MOSFETs, and are different value in the SOI layer and the bulk substrate. The intrinsic carrier concentration n_i is

$$n_i = n_{i0}T^{\frac{3}{2}} \exp\left(-\frac{E_g}{2q}\beta\right) \quad (27)$$

where E_g is the bandgap, and n_{i0} is the intrinsic carrier density at 300K.

The two types of surface potential, ϕ_s (front) and ϕ_b (back), are defined both at the location $y=0$ (source side) and at the pinch-off point (drain side) to give the following four surface potentials in all:

$$\phi_{s0} := \phi_s(y=0) \quad (28)$$

$$\phi_{b0} := \phi_b(y=0) \quad (29)$$

$$\phi_{sL} := \phi_s(y=L_{\text{eff}} - \Delta L) \quad (30)$$

$$\phi_{bL} := \phi_b(y=L_{\text{eff}} - \Delta L). \quad (31)$$

Note that the suffix “0” denotes the location $y=0$ (source side) and the suffix “L” denotes the drain-side pinch-off point. The pinch-off point is located at $y = L_{\text{eff}} - \Delta L$, where ΔL is the length of the pinch-off region under the saturation condition (see Section 9).

The surface potentials at the source side and the drain side are calculated independently with quasi-Fermi potential values ϕ_f . The electron quasi-Fermi potential $\phi_f(y)$ preserves the following relationship:

$$\phi_f(L_{\text{eff}} - \Delta L) - \phi_f(0) = V_{\text{ds,eff}} \quad (32)$$

where $\phi_f(0)$ is taken as zero. Please be reminded that the y -axis is defined along the direction of current flow (source is $y=0$), as was in Fig. 1.

The effective drain voltage $V_{\text{ds,eff}}$ is introduced to fit measured transition characteristics of the channel conductance g_{ds} between the linear region and the saturation region to compensate the charge-sheet approximation

$$V_{\text{ds,eff}} = \frac{V_{\text{ds}}}{\left[1 + \left(\frac{V_{\text{ds}}}{V_{\text{ds,sat}}}\right)^{\Delta}\right]^{\frac{1}{\Delta}}} \quad (33)$$

where

$$\Delta = \frac{\text{DDLTMAX} \cdot T1}{\text{DDLTMAX} + T1} + \text{DDLTICT} \quad (34)$$

$$T1 = \text{DDLTSPL} \cdot L_{\text{gate}}.^6 \quad (35)$$

$$V_{\text{ds,sat}} = V'_G + \frac{qN_{\text{subs}}\epsilon_{\text{Si}}}{C_{\text{FOX}}^2} \left[1 - \sqrt{1 + \frac{2C_{\text{FOX}}^2}{\beta qN_{\text{subs}}\epsilon_{\text{Si}}}(\beta V'_G - 1)} \right] \quad (36)$$

where V'_G is the same as (9) in this section.

As for charges, similar notations apply:

$$Q_{i0} := Q_i(y = 0) \quad (37)$$

$$Q_{b0} := Q_b(y = 0) \quad (38)$$

$$Q_{iL} := Q_i(y = L_{\text{eff}}) \quad (39)$$

$$Q_{bL} := Q_b(y = L_{\text{eff}}). \quad (40)$$

Note the suffix “i” stands for front gate inversion and “b” for back inversion. Or compactly,

$$Q_{x0} := Q_x(y = 0) \quad (41)$$

$$Q_{xL} := Q_x(y = L_{\text{eff}}) \quad (42)$$

where the suffix “x” denotes “i” (front) and “b” (back).

In deriving an equation for drain current and for charges, an integration of distributed charges along the channel is involved. Different from other HiSIM models, this integration is performed by approximating trapezoidal charge distributions as

$$\frac{1}{L_{\text{eff}}} \int_{y=0}^{y=L_{\text{eff}}} Q_x(y) dy = \frac{Q_x(y = 0) + Q_x(y = L_{\text{eff}})}{2} \quad (43)$$

where the suffix “x” denotes “i” (front) and “b” (back).

Charges and currents induced within the SOTB-MOSFET are connected to nodes as shown in Fig. 4.

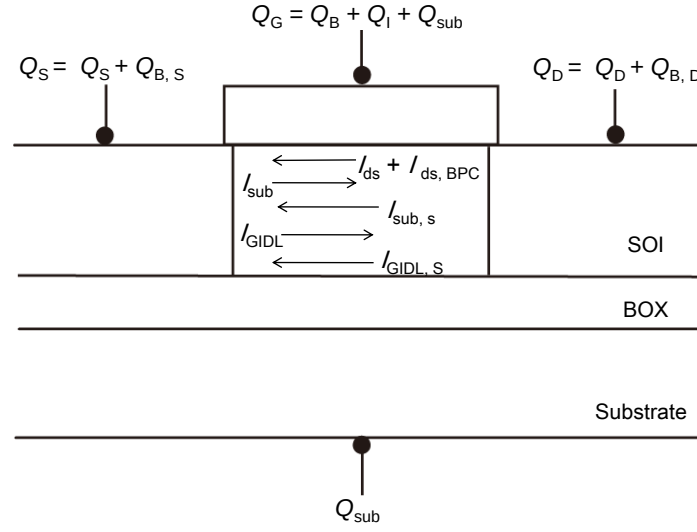


Fig. 4: Charges and currents induced within SOTB-MOSFET and connections to nodes. I_{ds} denotes drain current (Section 3); $I_{ds, BPC}$ denotes the current induced by the hole accumulation within the SOI layer (equation (230), Section 16.1); I_{sub} and $I_{sub, s}$ denote body current (Section 16.1); I_{GIDL} and $I_{GIDL, s}$ denote GID(S)L current (Section 16.3).

Table 2: HiSIM_SOTB model parameters introduced in Section 2 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

TFOX	front oxide thickness
TBOX	buried oxide thickness
TSOI	silicon film thickness
DDLTMAX	smoothing coefficient for V_{ds}
DDLTICT	L_{gate} -dependence of smoothing coefficient
DDLTSLP	L_{gate} -dependence of smoothing coefficient

3 Drain Current

Among charges induced within SOTB-MOSFET, two inversion charges, Q_i and Q_b (see Fig. 3), contribute to the current. Both current equations are derived by integrating corresponding charges. The final drain current observed is sum of these two currents, $I_{ds,f}$ and $I_{ds,b}$. Under the gradual-channel approximation with further approximations of an idealized gate structure and uniform channel doping, the drain current I_{ds} under the drift-diffusion approximation is written as [7] for the FD case

$$I_{ds} = I_{ds,f} + I_{ds,b} \quad (44)$$

$$I_{ds,f} = \frac{W_{\text{eff}} \cdot \mathbf{NF}}{(L_{\text{eff}} - \Delta L)} \cdot \mu_f \cdot \frac{I_{dd,f}}{\beta} \quad (45)$$

$$\text{where } I_{dd,f} = - \left[\beta \cdot \frac{(Q_{iL} + Q_{i0}) \cdot (\phi_{sL} - \phi_{s0})}{2} + Q_{iL} - Q_{i0} \right] \quad (46)$$

$$I_{ds,b} = \frac{W_{\text{eff}} \cdot \mathbf{NF}}{(L_{\text{eff}} - \Delta L)} \cdot \mu_b \cdot \frac{I_{dd,b}}{\beta} \quad (47)$$

$$\text{where } I_{dd,b} = - \left[\beta \cdot \frac{(Q_{bL} + Q_{b0}) \cdot (\phi_{bL} - \phi_{b0})}{2} + Q_{bL} - Q_{b0} \right] \quad (48)$$

In these equations, μ_f and μ_b are high-field carrier mobilities at the front inversion Q_i and the back inversion Q_b (see Eqs. 99 and 122 of Section 8), respectively.

In this section, no more model parameters have been introduced.

4 Threshold Voltage Shift

The surface-potential-based model requires no threshold voltage V_{th} in its core description of the model. However, HiSIM_SOTB has opted for modeling various submodels such as short-channel effects as a V_{th} shift from an otherwise ideal case.

The whole contribution to the ΔV_{th} is summarised as

$$\Delta V_{th} = \Delta V_{th,SC} + \Delta V_{th,R} + \Delta V_{th,P} + \Delta V_{th,SCR} + \Delta V_{th,W} + \Delta V_{th,sm} - \phi_{Spg} \quad (49)$$

where each term denotes:

- $\Delta V_{th,SC}$ denotes the contribution of the short-channel effects (equation 55, Section 4.1)
- $\Delta V_{th,R}$ denotes the contribution of additional short-channel effects in the presence of impurity concentration inhomogeneity in the vertical direction (retrograded implantation) (Section 4.2)
- $\Delta V_{th,P}$ denotes the contribution of the short-channel effects in the presence of impurity concentration inhomogeneity in the lateral direction (pocket implantation) (equation 56, Section 4.1)
- $\Delta V_{th,SCR}$ denotes the contribution of the short-channel effects which is specific to SOTB devices (equation 55, Section 4.1)
- $\Delta V_{th,W}$ denotes the contribution of the narrow-channel effects (equation 129, Section 10)
- $\Delta V_{th,sm}$ denotes the contribution of the small geometry effects (equation 154, Section 10.4)
- ϕ_{Spg} denotes the contribution of the poly-Si gate depletion effects (equation 74, Section 6)

The modeled ΔV_{th} is incorporated in the Poisson equation as can be seen in Eq. (9) and can be viewed as consisting of two main effects or components:

(I) Short-channel effect: $\Delta V_{th,SC}$

(II) Reverse-short-channel effect: $\Delta V_{th,R}$ and $\Delta V_{th,P}$

The separation into these two components ($\Delta V_{th} = \Delta V_{th,SC} + \Delta V_{th,R}$ (or $\Delta V_{th,P}$)) is schematically shown in Fig. 5.

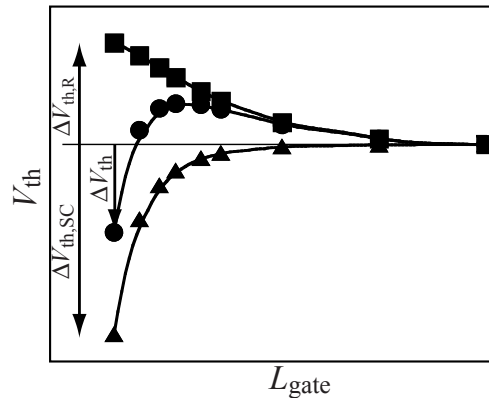


Fig. 5: Schematic plot of the separation of V_{th} into the contributions of the short-channel and the reverse-short-channel effect.

4.1 (I) Short-Channel Effects

By applying the Gauss's law, $\Delta V_{th,SC}$ can be written as a function of the lateral electric field E_y . A parabolic potential distribution along the channel is approximated, which results in a position-independent gradient of the lateral electric field $\frac{dE_y}{dy}$ [4]

$$\Delta V_{th,SC} = \frac{\epsilon_{Si}}{C_{FOX}} W_d \frac{dE_y}{dy} \quad (50)$$

W_d is the depletion-layer thickness, which is fixed to **TSOI**, and $\frac{dE_y}{dy}$ is derived with model parameters in the form

$$\frac{dE_y}{dy} = \frac{2(\mathbf{VBI} - 2\Phi'_B)}{(L_{gate} - \mathbf{PARL2})^2} \left(\mathbf{SC1} + \mathbf{SC2} \cdot V_{ds} + \mathbf{SC3} \cdot \frac{2\Phi_B - V_{es}}{L_{gate}} + \mathbf{SC5} \cdot V_{es} \right) \quad (51)$$

where

$$\Phi'_B = \Phi_B + \mathbf{PTHROU} \cdot (\Phi''_B(V_{gs}) - \Phi_B) \quad (52)$$

and Φ''_B is the surface potential calculated analytically by solving the Poisson equation under the diffusion approximation

$$\Phi''_B(V_{gs}) = V_G + \left(\frac{\text{const0}}{C_{FOX}} \frac{\beta}{2} \right)^2 \left[1 + \sqrt{1 - \frac{4\beta(V_G - V_{es}) - 4}{\beta^2 \left(\frac{\text{const0}}{C_{FOX}} \right)^2}} \right] \quad (53)$$

where

$$V_G = V_{gs} - \mathbf{VFBC} + \Delta V_{th,SCR} \quad (54)$$

VBI and **PARL2** represent the built-in potential and the depletion width of the junction lateral to the channel, respectively. The model parameter **SC1** determines the threshold voltage shift for small V_{ds} and V_{es} , and is expected to be unity. If measured V_{th} is plotted as a function of V_{ds} , it shows nearly a linear dependence. The gradient of measured V_{th} as a function of V_{ds} is **SC2**. **SC3** is a fitting parameter describing the V_{es} dependence of V_{th} . These fitting parameters are introduced to avoid iteration procedure needed for the whole calculation procedure by adjusting them to measured threshold voltage shift. **PTHROU** describes the increase of the subthreshold swing for short-channel transistors.

In SOTB-MOSFET, additional short channel effect is observed because of the electric field via BOX. This effect is modeled as follows [9]:

$$\Delta V_{th,SCR} = \frac{\mathbf{SCR1} \cdot \mathbf{TFOX} (E_g + 2\Phi_B - \mathbf{SCR3} + \mathbf{SCR2} \cdot V_{ds})}{\frac{L_{gate}}{2} + \mathbf{PARL1}} \quad (55)$$

where E_g is the bandgap determined in Section 13.

4.2 (II) Reverse-Short-Channel Effects

The reverse-short-channel effect is categorized into resulting from two physical MOSFET properties:

- (i) **Impurity concentration inhomogeneity in the direction vertical to the channel (vertical channel inhomogeneity)** (obvious in the retrograded implantation): $\Delta V_{th,R}$
- (ii) **Impurity concentration inhomogeneity in the direction parallel to the channel (lateral channel inhomogeneity)** (obvious in the pocket implantation): $\Delta V_{th,P}$

(i) **Impurity concentration inhomogeneity in the direction vertical to the channel (Retrograded Implantation)**

The model part described in section 4.2. (i) is not included in HiSIM_SOTB explicitly, but the above model parameters **SC3** (see above) and **SCP3** (see below) can be used to describe the effect successfully, if the inhomogeneity is not extremely large.

(ii) Impurity concentration inhomogeneity in the lateral direction parallel to the channel (Pocket Implantation)

The pocket-implant technology causes an inhomogeneity along the channel. Two model parameters (**LP**: length of the pocket extension into the channel; **NSUBP**: peak of the pocket impurity concentration) are introduced. The impurity concentration in the SOI layer is distinguished with **NSUBS** from **NSUBP** [1, 8]. The resulting model equations for the V_{th} shift due to the pocket implant are:

$$\Delta V_{th,P} = (V_{th,R} - V_{th0}) \frac{\epsilon_{Si}}{C_{FOX}} \mathbf{TSOI} \frac{dE_{y,P}}{dy} \quad (56)$$

$$V_{th,R} = V_{FB} + 2\Phi_B + \frac{Q_{B0}}{C_{FOX}} + P_{tovr} \quad (57)$$

$$Q_{B0} = \sqrt{2q \cdot N_{subs} \cdot \epsilon_{Si} \cdot (2\Phi_B - V_{es})} \quad (58)$$

$$V_{th0} = V_{FB} + 2\Phi_{BC} + \frac{\sqrt{2q N_{subsp} \epsilon_{Si} (2\Phi_{BC} - V_{es})}}{C_{FOX}} \quad (59)$$

$$\frac{dE_{y,P}}{dy} = \frac{2(\mathbf{VBI} - 2\Phi_B)}{\mathbf{LP}^2} \left(\mathbf{SCP1} + \mathbf{SCP2} \cdot V_{ds} + \mathbf{SCP3} \cdot \frac{2\Phi_B - V_{es}}{\mathbf{LP}} \right) \quad (60)$$

$$P_{tovr} = \begin{cases} \frac{1}{\beta} \ln \left(\frac{N_{subb0}}{N_{subsp}} \right) & (L_{gate} \leq 2 \cdot \mathbf{LP}) \\ 0 & (L_{gate} > 2 \cdot \mathbf{LP}) \end{cases} \quad (61)$$

$$N_{subp0} = 2 \cdot N_{subps} - \frac{(N_{subps} - N_{subsp}) \cdot L_{gate}}{\mathbf{LP}} - N_{subsp} \quad (62)$$

$V_{th,R}$ and V_{th0} , defined in Eqs. (57) and (59), are the threshold voltages for the cases with and without pocket-implant, respectively. The parameters **SCP1** – **SCP3** describe the short-channel effect caused by the potential minimum at the higher impurity concentration of the pocket. $2\Phi_{BC}$ is the potential giving threshold condition with **NSUBS** and $2\Phi_B$ is the equivalent potential with N_{subs} as follows:

$$\Phi_{BC} = \frac{2}{\beta} \ln \left(\frac{N_{subs}}{n_i} \right) \quad (63)$$

$$\Phi_B = \frac{2}{\beta} \ln \left(\frac{N_{subsp}}{n_i} \right) \quad (64)$$

$$N_{subs} = \begin{cases} \frac{N_{subsp}(L_{gate} - \mathbf{LP}) + N_{subps} \cdot \mathbf{LP}}{L_{gate}} & (L_{gate} > \mathbf{LP}) \\ N_{subps} + \frac{(N_{subps} - N_{subsp}) \cdot (\mathbf{LP} - L_{gate})}{\mathbf{LP}} & (L_{gate} \leq \mathbf{LP}) \end{cases} \quad (65)$$

As defined in Eq. (65), N_{subs} is replaced to the averaged impurity concentration in the channel and N_{subps} is introduced, beginning from channel lengths where pockets at source and drain start to overlap.

4.3 Lateral inhomogeneity of the substrate impurity concentration N_{SUBB}

Similar to the pocket implantation, the substrate impurity concentration along the lateral direction is not homogeneously distributed. This causes the channel length dependent N_{SUBB} modeled as

$$N_{subb} = \mathbf{NSUBB} \cdot \left(1 + \frac{\mathbf{NSUBBL}}{(L_{gate} \cdot 10^6) \mathbf{NSUBBLP}} \right) \quad (66)$$

4.4 Flat-Band Voltage Shift due to Mechanical Stress

It is known that the stress induced on the gate changes the work function value, which causes the flat-band shift. This effect is introduced as

$$V_{fb} = f(V_{fb1}, V_{fb2}, V_{fb3}) \quad (67)$$

where f means a function. In this function, V_{fb1} is smoothly limited not to exceed V_{fb2} , and the resultant value is also smoothly limited not to exceed V_{fb3} . Those intermediate values V_{fb1} , V_{fb2} and V_{fb3} are introduced for the modeling of the

gate-length dependence in the following expressions

$$V_{fb1} = VFBC \cdot \left(1 + \frac{VFBCL1}{(L_{gate} \cdot 10^6) VFBCL1P} \right) \quad (68)$$

$$V_{fb2} = VFBC \cdot \left(1 + \frac{VFBCL2}{(L_{gate} \cdot 10^6) VFBCL2P} \right) \quad (69)$$

$$V_{fb3} = VFBC + VFBHAMP \cdot (L_{gate} \cdot 10^6) \quad (70)$$

Note that V_{fb2} is a limitation on maximum value of V_{fb1} , although these two equations are symmetrical; and V_{fb3} can become huge value when L_{gate} is long.

Table 3: HiSIM_SOTB model parameters introduced in Section 4 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

VBI	built-in potential
PARL1	SOI SCE parameter
PARL2	depletion width of channel/contact junction
SC1	magnitude of short-channel effect
SC2	V_{ds} -dependence of short-channel effect
*SC3	V_{es} -dependence of short-channel effect
SC5	V_{es} -dependence of short-channel effect
PTHROU	modify subthreshold slope
SCR1	parameter for SCE via BOX
SCR2	parameter for SCE via BOX
SCR3	parameter for SCE via BOX
SCP1	magnitude of short-channel effect due to pocket
SCP2	V_{ds} -dependence of SCE due to pocke
*SCP3	V_{es} -dependence of SCE due to pocke
LP	pocket penetration length
NSUBB	substrate impurity concentration
NSUBBL	channel-length dependence of NSUBB
NSUBBLP	channel-length dependence of NSUBB
NSUBP	max pocket concentration
VFBC	flat-band voltage
VFBCL1	channel length dependence of flat-band voltage
VFBCL1P	channel length dependence of flat-band voltage
VFBCL2	channel length dependence of flat-band voltage
VFBCL2P	channel length dependence of flat-band voltage
VFBHAMP	channel length dependence of flat-band voltage

5 Punchthrough

Origin of the punchthrough effect is the bipolar effect through source, substrate, and drain. The effect is described by a power function of the potential difference instead of the exponential function. The current contributed by this mechanism is denoted by I_{PUNCH} and expressed as:

$$I_{\text{punch}} = I_{\text{PUNCH,f}} + I_{\text{PUNCH,b}}$$

$$I_{\text{PUNCH,f}} = \frac{W_{\text{eff}} \cdot \mathbf{NF}}{L_{\text{eff}}} \frac{\mu}{\beta} \cdot (\phi_{\text{sL}} - \phi_{\text{s0}}) \left\{ C_{\text{FOX}} \cdot \beta \frac{\mathbf{PTL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{PTLP}}} \cdot (V_{\text{bi}} - \phi_{\text{s0}})^{\mathbf{PTP}} \cdot \left(1 + \mathbf{PT2} \cdot V_{\text{ds}} + \frac{\mathbf{PT4} \cdot (\phi_{\text{s0}} - V_{\text{es}})}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{PT4P}}} \right) + \text{COND} \right\} \quad (71)$$

$$I_{\text{PUNCH,b}} = \frac{W_{\text{eff}} \cdot \mathbf{NF}}{L_{\text{eff}}} \frac{\mu}{\beta} \cdot (\phi_{\text{bL}} - \phi_{\text{b0}}) \left\{ C_{\text{FOX}} \cdot \beta \frac{\mathbf{PTL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{PTLP}}} \cdot (V_{\text{bi}} - \phi_{\text{b0}})^{\mathbf{PTP}} \cdot \left(1 + \mathbf{PT2} \cdot V_{\text{ds}} + \frac{\mathbf{PT4} \cdot (\phi_{\text{b0}} - V_{\text{es}})}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{PT4P}}} \right) + \text{COND} \right\} \quad (72)$$

$$V_{\text{bi}} = 1.1$$

The high field under the saturation condition causes the pinch-off region and the current flows away from the surface. This effect is considered as the lateral-field-induced charge for the capacitance (see Section 15). The simplified formulation is applied to consider the effect as

$$\text{COND} = C_{\text{FOX}} \cdot \beta \frac{\mathbf{GDL}}{(L_{\text{gate}} \cdot 10^6 + \mathbf{GDLD} \cdot 10^6)^{\mathbf{GDLP}}} \cdot V_{\text{ds}} \quad (73)$$

This effect is considered only for the front current.

Table 4: HiSIM_SOTB model parameters introduced in Section 5 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

front/back	
PTL	strength of front/back current
PTLP	L_{gate} dependence of front/back current
PTP	strength of front/back current
PT2	V_{ds} dependence of front/back current
PT4	V_{es} dependence of front/back current
PT4P	V_{es} dependence of front/back current
front	
* GDL	strength of high-field effect in front current
* GDLP	L_{gate} dependent high-field effect in front current
* GDLD	L_{gate} dependent high-field effect in front current

6 Depletion Effect of the Gate Poly-Si

To simplify the calculation procedure for the circuit-simulation application, the poly-depletion effect, the potential drop within the poly-Si ϕ_{SpG} , is approximated as a function of V_{gs} as shown in Eq. (74), and is included in the ΔV_{th} calculation as a potential drop of V_{gs} [10]

$$\phi_{\text{SpG}} = \text{PGD1} \left(1 + \frac{1}{L_{\text{gate}} \cdot 10^6} \right)^{\text{PGD4}} \exp \left(\frac{V_{\text{gs}} - \text{PGD2}}{[1V]} \right) \quad (74)$$

In Eq. (74), **PGD1** describes the strength of the poly-depletion, **PGD2** represents the threshold voltage of the poly-depletion, and **PGD4** is introduced to describe the channel length dependence. The function of Eq. (74) is smoothed so that ϕ_{SpG} does not exceed ϕ_{S0} for very large V_{gs} . [1V] in the exp term is introduced just to delete dimension.

Table 5: HiSIM_SOTB model parameters introduced in Section 6 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

PGD1	strength of poly-depletion effect
PGD2	threshold voltage of poly-depletion effect
*PGD4	L_{gate} -dependence of poly-depletion effect

7 Quantum-Mechanical Effects

The main quantum-mechanical phenomenon, which has to be included into MOSFET model for circuit simulation, is the repulsion of the channel's carrier-density peak into the substrate away from the surface. This can be described phenomenologically by an increased effective oxide thickness ΔT_{FOX} .

After some simplifications final equations implemented into HiSIM_SOTB for the reproduction of quantum mechanical effects are:

$$T_{\text{FOX}} = \text{TFOX} + \Delta T_{\text{FOX}} \quad (75)$$

$$\Delta T_{\text{FOX}} = \frac{\text{QME1}}{V_{\text{gs}} - V_{\text{th}}(T_{\text{FOX}} = \text{TFOX}) - \text{QME2}} + \text{QME3} \quad (76)$$

where **QME1**, **QME2**, and **QME3** are the quantum-effect model parameters.

Table 6: HiSIM_SOTB model parameters introduced in Section 7 of this manual.

QME1	V_{gs} -dependence of quantum mechanical effect
QME2	V_{gs} -dependence of quantum mechanical effect
QME3	minimum T_{FOX} modification

8 Mobility Model

8.1 Front Side Mobility

As for the low-field mobility the universal relationship is still applicable for the SOTB-MOSFET generation, which consists of three independent scattering mechanisms of Coulomb, phonon and surface-roughness scattering [11]:

$$\frac{1}{\mu_0} = \frac{1}{\mu_{CB}} + \frac{1}{\mu_{PH}} + \frac{1}{\mu_{SR}} \quad (77)$$

$$\mu_{CB}(\text{Coulomb}) = M_{\text{Coulomb}0} + M_{\text{Coulomb}1} \cdot \frac{|Q_I|}{q \times 10^{11}} \quad (78)$$

$$\mu_{PH}(\text{phonon}) = \frac{M_{uephonon}}{E_{eff}^{MUEPH0}} \quad (79)$$

$$\mu_{SR}(\text{surface roughness}) = \frac{MUESR1}{E_{eff}^{Muesurface}} \quad (80)$$

Here E_{eff} is the effective field normal to the surface and $\mu_{PH}(\text{phonon})$ is modeled temperature dependent according to Eq. (179) in Section 13.

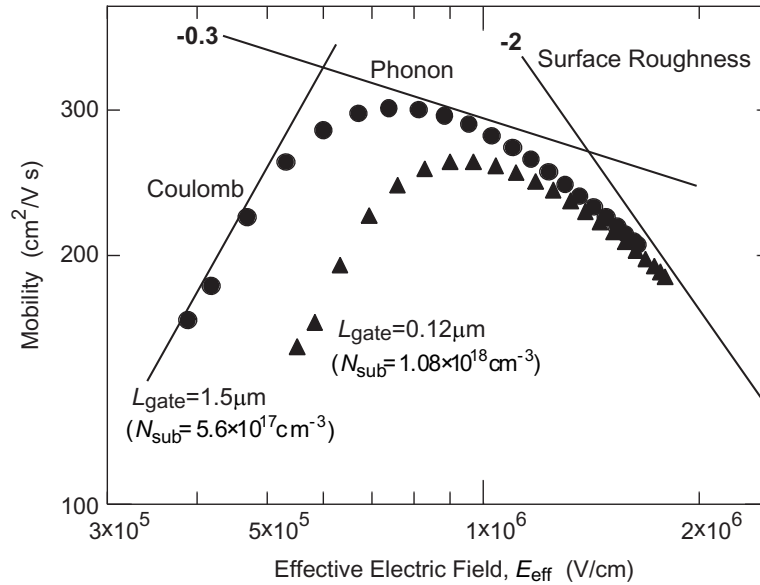


Fig. 6: Calculated mobility as a function of effective field for different MOSFET devices.

When the effective field is low (less than about $5 \cdot 10^5$ V/cm), the Coulomb scattering dominates the mobility (see Fig. 6). Hence we can introduce a minimum value of E_{eff} as follows:

$$E_{eff} = \max(E_{eff}, 3 \cdot 10^3) \quad (81)$$

The flag **CONEWMUB** is introduced to preserve old model cards.

$$E_{eff} = \begin{cases} E_{eff0} & (\text{when } \mathbf{CONEWMUB} = 0) \\ E_{eff0} + \frac{\epsilon_{ox}}{\epsilon_{si}} \cdot \frac{\phi_b - \phi_{bulk} - V'_{es}}{T_{BOX}} & (\text{when } \mathbf{CONEWMUB} = 1) \end{cases} \quad (82)$$

where V'_{es} includes the built-in potential between the SOI layer and the bulk substrate.

E_{eff0} is expressed as:

$$E_{\text{eff0}} = \frac{1}{\epsilon_{\text{Si}}} (N_{\text{dep}} \cdot Q_{\text{s,dep}} + N_{\text{inv}} \cdot Q_{\text{IB}}) \left(\frac{1}{1 + P_{\text{ds2}} \cdot \mathbf{NINVD}} \right) \quad (83)$$

$$Q_{\text{IB}} = Q_{\text{I}} - MUE_{\text{qb}} \cdot Q_{\text{b}}(L) \quad (84)$$

$$N_{\text{dep}} = \mathbf{NDEP} \cdot \left(1 + \frac{\mathbf{NDEPL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{NDEPLP}}} \right) \quad (85)$$

$$N_{\text{inv}} = \mathbf{NINV} \cdot \left(1 + \frac{\mathbf{NINVL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{NINVLP}}} \right) \quad (86)$$

$$P_{\text{ds2}} = (\phi_{\text{sL}} - \phi_{\text{s0}})^{\mathbf{NINVDP}} \quad (87)$$

where

$$Q_{\text{s,dep}} = \frac{Q_{\text{s,dep}}(L) + Q_{\text{s,dep}}(0)}{2} \quad (88)$$

$$Q_{\text{I}} = \frac{Q_{\text{i}}(L) + Q_{\text{i}}(0)}{2} \quad (89)$$

$$MUE_{\text{qb}} = \mathbf{MUEQB} \cdot \left(1 + \frac{\mathbf{MUEQBL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{MUEQBLP}}} \right) \quad (90)$$

Note that $L_{\text{gate}} \cdot 10^6$ becomes a unitless quantity.

The mobility universality preserves following conditions [12, 13]

$$\mathbf{MUEPH0} \simeq 0.3 \quad (91)$$

$$M_{\text{uesurface}} = 2.0 \quad (92)$$

$$\mathbf{NDEP} = 1.0 \quad (93)$$

$$\mathbf{NINV} = 0.5 \quad (94)$$

independent of technology variations. However, these parameters can be used for fitting purposes [14]. It is expected that the inversion condition is not so strong as the bulk MOSFET because the electric field induced by the front gate is weakened by that of the back gate.

Due to the carrier flow at increasing distance from the surface with reducing L_{gate} , the electric field experienced by the carriers is different from the field in the long L_{gate} case. This results in a modification of M_{uephonon} , which is modeled as

$$M_{\text{uephonon}} = \mathbf{MUEPH1} \cdot \left(1 + \frac{\mathbf{MUEPHL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{MUEPLP}}} \right) \quad (95)$$

The surface-roughness model parameter $\mathbf{MUESR0}$ is modeled to have a similar channel length dependence written as

$$M_{\text{uesurface}} = \mathbf{MUESR0} \cdot \left(1 + \frac{\mathbf{MUESRL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{MUESLP}}} \right) \quad (96)$$

M_{uephonon} and $M_{\text{uesurface}}$ can be modified due to mechanical stress or small geometry as explained in Section 10.

The channel dependence of the screening effect of the Coulomb scattering is introduced as

$$M_{\text{Coulomb0}} = \mathbf{MUECB0} \cdot (L_{\text{gate}} \cdot 10^6)^{\mathbf{MUECB0LP}} \cdot \left(1 + \frac{\mathbf{MUECB0L2}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{MUECB0L2P}}} \right) \quad (97)$$

$$M_{\text{Coulomb1}} = \mathbf{MUECB1} \cdot (L_{\text{gate}} \cdot 10^6)^{\mathbf{MUECB1LP}} \cdot \left(1 + \frac{\mathbf{MUECB1L2}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{MUECB1L2P}}} \right) \quad (98)$$

The high-field mobility is modeled as [15]

$$\mu_f = \frac{\mu_0}{\left(1 + \left(\frac{\mu_0 E_y}{V_{\max}}\right)^{\mathbf{BB}}\right)^{\frac{1}{\mathbf{BB}}}} \quad (99)$$

$$E_y = \sqrt{\left(\frac{I_{\text{dd},f}}{\beta \cdot Q_i(0) \cdot L_{\text{eff}}}\right)^2 + \left(\frac{0.2 \cdot V_{\max}}{\mu_0}\right)^2} \quad (100)$$

where the maximum velocity $V_{\max}$ is modeled temperature dependent, as described in Section 13. $\mathbf{BB}$ is usually fixed to 2 for electrons and 1 for holes. E_y is the lateral electric field induced within the device.

The maximum velocity $V_{\max}$ should be the maximum electron-saturation velocity ($\simeq 1 \times 10^7$ cm/s), which is however exceeded at reduced L_{gate} . This phenomenon, called velocity overshoot, is included in the mobility model in the following manner

$$V_{\max} = \mathbf{VMAX} \cdot \left(1 + \frac{\mathbf{VOVER}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{VOVERP}}}\right) \cdot \left(1 + \frac{\mathbf{VOVERL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{VOVERLP}}}\right) \quad (101)$$

8.2 Back Side Mobility

As for the low-field mobility the universal relationship is still applicable for the SOTB-MOSFET generation, which consists of three independent scattering mechanisms of Coulomb, phonon and surface-roughness scattering [11]:

$$\frac{1}{\mu_0} = \frac{1}{\mu_{\text{CB}}} + \frac{1}{\mu_{\text{PH}}} + \frac{1}{\mu_{\text{SR}}} \quad (102)$$

$$\mu_{\text{CB}}(\text{Coulomb}) = M_{\text{Coulomb0b}} + M_{\text{Coulomb1b}} \cdot \frac{|Q_{\text{B}}|}{q \times 10^{11}} \quad (103)$$

$$\mu_{\text{PH}}(\text{phonon}) = \frac{M_{\text{uephononb}}}{E_{\text{eff}}^{\mathbf{MUEPH0B}}} \quad (104)$$

$$\mu_{\text{SR}}(\text{surface roughness}) = \frac{\mathbf{MUESR1B}}{E_{\text{eff}}^{\mathbf{Muesurfaceb}}} \quad (105)$$

Here E_{eff} is the effective field normal to the surface and $\mu_{\text{PH}}(\text{phonon})$ is modeled temperature dependent according to Eq. (179) in Section 13.

When the effective field is low (less than about 10^5 V/cm), the Coulomb scattering dominate the mobility (see Fig. 6). Hence we can introduce a minimum value of E_{eff} as follows:

$$E_{\text{eff}} = \max(E_{\text{eff}}, 3 \cdot 10^1) \quad (106)$$

The flag **CONEWMUB** is introduced to preserve old model cards.

$$E_{\text{eff}} = \begin{cases} E_{\text{eff0}} & (\text{when } \mathbf{CONEWMUB} = 0) \\ \frac{\epsilon_{\text{ox}}}{\epsilon_{\text{Si}}} \cdot \frac{\phi_{\text{b}} - \phi_{\text{bulk}}}{T_{\text{BOX}}} & (\text{when } \mathbf{CONEWMUB} = 1) \end{cases} \quad (107)$$

$$E_{\text{eff0}} = \frac{1}{\epsilon_{\text{Si}}} (N_{\text{dep}} \cdot Q_{\text{b,dep}} + N_{\text{inv}} \cdot Q_{\text{IB}}) \left(\frac{1}{1 + P_{\text{dsb2}} \cdot \mathbf{NINVD}} \right) \quad (108)$$

$$Q_{\text{IB}} = Q_{\text{B}} - MUE_{\text{qb}} \cdot Q_i(L) \quad (109)$$

$$P_{\text{dsb2}} = (\phi_{\text{bL}} - \phi_{\text{b0}})^{\mathbf{NINVDP}} \quad (110)$$

where

$$Q_{b,dep} = \frac{Q_{b,dep}(L) + Q_{b,dep}(0)}{2} \quad (111)$$

$$Q_B = \frac{Q_b(L) + Q_b(0)}{2} \quad (112)$$

$$Q_I = \frac{Q_i(L) + Q_i(0)}{2} \quad (113)$$

$$MUE_{qb} = MUE_{qbb} \cdot \left(1 + \frac{MUE_{qbl}}{(L_{gate} \cdot 10^6)^{MUE_{qblp}}} \right) \quad (114)$$

$$(115)$$

The mobility universality preserves following conditions [12, 13]

$$MUE_{PH0B} \simeq 0.3 \quad (116)$$

$$M_{uesurfaceb} = 2.0 \quad (117)$$

independent of technology variations. However, these parameters can be used for fitting purposes [14]. Due to the carrier flow at increasing distance from the surface with reducing L_{gate} , the electric field experienced by the carriers is different from the field in the long L_{gate} case. This results in a modification of $M_{uephonon}$, which is modeled as

$$M_{uephononb} = MUE_{PH1B} \cdot \left(1 + \frac{MUE_{PHLB}}{(L_{gate} \cdot 10^6)^{MUE_{PLPB}}} \right) \quad (118)$$

The surface-roughness model parameter $MUESR0B$ is modeled to have a similar channel length dependence written as

$$M_{uesurfaceb} = MUESR0B \cdot \left(1 + \frac{MUESRLB}{(L_{gate} \cdot 10^6)^{MUESLPB}} \right) \quad (119)$$

$M_{uephononb}$ and $M_{uesurfaceb}$ can be modified due to mechanical stress or small geometry as explained in Section 10.

The channel dependence of the screening effect of the Coulomb scattering is introduced as

$$M_{Coulomb0b} = MUECB0B \cdot (L_{gate} \cdot 10^6)^{MUECB0LPB} \cdot \left(1 + \frac{MUECB0L2B}{(L_{gate} \cdot 10^6)^{MUECB0L2PB}} \right) \quad (120)$$

$$M_{Coulomb1b} = MUECB1B \cdot (L_{gate} \cdot 10^6)^{MUECB1LPB} \cdot \left(1 + \frac{MUECB1L2B}{(L_{gate} \cdot 10^6)^{MUECB1L2PB}} \right) \quad (121)$$

The high-field mobility is modeled as [15]

$$\mu_b = \frac{\mu_0}{\left(1 + \left(\frac{\mu_0 E_y}{V_{max}} \right)^{BB} \right)^{\frac{1}{BB}}} \quad (122)$$

$$E_y = \sqrt{\left(\frac{I_{dd,b}}{\beta \cdot Q_b(0) \cdot L_{eff}} \right)^2 + \left(\frac{0.2 \cdot V_{max}}{\mu_0} \right)^2} \quad (123)$$

where the maximum velocity V_{max} is modeled temperature dependent, as described in Section 13. BB is usually fixed to 2 for electrons and 1 for holes. E_y is the lateral electric field induced within the device.

Table 7: HiSIM.SOTB model parameters introduced in Section 8 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

front MUECB0 MUECB0LP MUECB0L2 MUECB0L2P MUECB1 MUECB1LP MUECB1L2 MUECB1L2P MUEPH0 MUEPH1 *MUEPHL *MUEPLP MUESR0 MUESR1 *MUESRL *MUESLP MUEQB	Coulomb scattering for current length dependence of Coulomb scattering for current length dependence of Coulomb scattering for current length dependence of Coulomb scattering for current Coulomb scattering for current length dependence of Coulomb scattering for current length dependence of Coulomb scattering for current length dependence of Coulomb scattering for current phonon scattering for current phonon scattering for current length dependence of phonon mobility reduction for current length dependence of phonon mobility for current surface-roughness scattering for current surface-roughness scattering for current length dependence of surface roughness mobility reduction for current length dependence of surface roughness mobility reduction for current magnitude of Q_{BL} for front side mobility
back MUECB0B MUECB0LPB MUECB0L2B MUECB0L2PB MUECB1B MUECB1LPB MUECB1L2B MUECB1L2PB MUEPH0B MUEPH1B *MUEPHLB *MUEPLPB MUESR0B MUESR1B *MUESRLB *MUESLPB MUEQBB	Coulomb scattering for current length dependence of Coulomb scattering for back current length dependence of Coulomb scattering for back current length dependence of Coulomb scattering for back current Coulomb scattering for back current length dependence of Coulomb scattering for back current length dependence of Coulomb scattering for back current length dependence of Coulomb scattering for back current phonon scattering for current phonon scattering for current length dependence of phonon mobility for current length dependence of phonon mobility for current surface-roughness scattering for current surface-roughness scattering for current length dependence of surface roughness mobility for current length dependence of surface roughness mobility for current magnitude of Q_{BL} for back side mobility
NDEP *NDEPL *NDEPLP NINV *NINVD *NINVDP BB VMAX VOVER VOVERP VOVERL VOVERLP MUEQBL MUEQBLP	depletion charge contribution on effective-electric field modify Q_B contribution for short channel modify Q_B contribution for short channel inversion charge contribution on effective-electric field reduced resistance effect for small V_{ds} reduced resistance effect for small V_{ds} high-field-mobility degradation maximum saturation velocity velocity overshoot effect L dependence of velocity overshoot velocity overshoot effect L dependence of velocity overshoot length dependence of MUEQB and MUEQBB length dependence of MUEQB and MUEQBB

9 Channel-Length Modulation

This effect is modeled for the front current, which is the dominating current under the strong inversion condition. The strength of the channel-length modulation is dependent on the junction profile between the channel and the drain contact. This dependence is modeled with the parameter **CLM1** as

$$\phi_S(\Delta L) = (1 - \mathbf{CLM1}) \cdot \phi_{sL} + \mathbf{CLM1} \cdot (\phi_{s0} + V_{ds}) \quad (124)$$

where **CLM1** can be interpreted to represent the abruptness of the junction and must be in the range $0 \leq \mathbf{CLM1} \leq 1$. Here **CLM1** = 1 means that the contact profile is abrupt and the complete potential increase occurs within the pinch-off region, whereas **CLM1** = 0 corresponds to the opposite condition and there is no steep potential increase in the pinch-off region but the most potential increase occurs within the overlap region.

The length of the pinch-off region ΔL is derived as

$$\Delta L = \frac{1}{2} \left[-\frac{1}{L_{\text{eff}}} \left(2 \frac{I_{dd,f}}{\beta Q_i} z + 2 \frac{q N_{\text{subs}}}{\epsilon_{\text{Si}}} (\phi_s(\Delta L) - \phi_{sL}) z^2 + E_0 z^2 \right) + \sqrt{\frac{1}{L_{\text{eff}}^2} \left(2 \frac{I_{dd,f}}{\beta Q_i} z + 2 \frac{q N_{\text{subs}}}{\epsilon_{\text{Si}}} (\phi_s(\Delta L) - \phi_{sL}) z^2 + E_0 z^2 \right)^2 + 4 \left(2 \frac{q N_{\text{sub}}}{\epsilon_{\text{Si}}} (\phi_s(\Delta L) - \phi_{sL}) z^2 + E_0 z^2 \right)} \right] \quad (125)$$

where E_0 is fixed to 10^5 and an auxiliary parameter z is

$$z = \frac{\epsilon_{\text{Si}} W_d}{CL_{M2} \cdot Q_b + \mathbf{CLM3} \cdot Q_i} \quad (126)$$

where

$$CL_{M2} = \begin{cases} \frac{5.0 \cdot 10^9}{\mathbf{TSOI_NSUBS}} & (\text{default}) \\ \mathbf{CLM2} & (\text{if } \mathbf{CLM2} \text{ is given in the model card.}) \end{cases} \quad (127)$$

and $I_{dd,f}$ is the quantity I_{dd} introduced in Section 3, herewith specifically for the front-gate side.

Two model parameters **CLM2** and **CLM3** are introduced to consider the uncertainty of Q_i in the pinch-off region and to counterbalance the two contributions from $Q_{\text{dep},\text{SOI}} (= q N_{\text{sub}} T_{\text{SOI}})$ and Q_i . It has to be notified that ΔL is equal to zero, when **CLM1** = 0. Though ΔL is determined mostly by $\phi_s(\Delta L)$, the combination between **CLM2** and **CLM3** influences the lateral-field-induced capacitance C_{Q_y} described in Section 15.

For the pocket implantation the lateral field increase is observed even for long-channel transistors. The effect is modeled by the threshold voltage shift. However, the degradation of the channel conductance under the saturation condition is much stronger than that described by the threshold voltage shift alone. This additional pocket effect is modeled as

$$\Delta L \leftarrow \Delta L (1 + \mathbf{CLM6} \cdot (L_{\text{gate}} \cdot 10^6)^{\mathbf{CLM5}}) \quad (128)$$

It can happen that $L_{\text{eff}} - \Delta L$ becomes negative, if extracted **CLM5** and **CLM6** values are out of acceptable ranges. In this case HiSIM_SOTB gives “warning” and fixes $L_{\text{eff}} - \Delta L$ to 1 nm.

Table 8: HiSIM_SOTB model parameters introduced in Section 9 of this manual.

CLM1	abruptness coefficient of channel/contact junction
CLM2	coefficient for Q_B contribution
CLM3	coefficient for Q_I contribution
CLM5	CLM parameter
CLM6	CLM parameter

10 Narrow-Channel Effects

10.1 Threshold Voltage Modification

The V_{th} reduction for reduced channel width (W_{gate}) is modeled under inclusion of the edge-fringing capacitances C_{ef} at the edge of the trench [16] as

$$\Delta V_{th,W} = \left(\frac{1}{C_{FOX}} - \frac{1}{C_{FOX} + 2C_{ef}/(L_{eff}W_{eff})} \right) \cdot Q_{dep,SOI} + \frac{\mathbf{WVTH0}}{W_{gate} \cdot 10^6} \quad (129)$$

where **WVTH0** is the parameter for including the basic width dependence and

$$C_{ef} = \frac{2\epsilon_{ox}}{\pi} L_{eff} \ln \left(\frac{2 \cdot T_{oxiso}}{T_{FOX}} \right) \quad (130)$$

$$= \frac{\mathbf{WFC}}{2} L_{eff} \quad (131)$$

where T_{oxiso} is an oxide effective thickness at the device isolation edge, which value would not be well determined. Therefore, an alternative expression (equation (131)) is provided with a model parameter. Here, **WFC** is the model parameter for including the edge-fringing-capacitance effects.

Enhancement of the V_{th} reduction with decreasing W_{gate} is often measured when L_{gate} is also small. This effect becomes obvious for the pocket technology and is modeled by a modification of the pocket impurity concentration as

$$N_{subpp} = \mathbf{NSUBP} \cdot \left(1 + \frac{\mathbf{NSUBP0}}{(W_{gate} \cdot 10^6)^{\mathbf{NSUBWP}}} \right) \quad (132)$$

The width dependence of the substrate impurity concentration N_{subs} is also considered as

$$N_{subsp} = \min \left(\mathbf{NSUBSMAX}, \mathbf{NSUBS} \cdot \left(1 + \frac{\mathbf{NSUBSW}}{(W_{gate} \cdot 10^6)^{\mathbf{NSUBSWP}}} \right) \right) \quad (133)$$

The width dependent substrate impurity concentration is considered as

$$N_{subbl} = \mathbf{NSUBB} \cdot \left(1 + \frac{\mathbf{NSUBBW}}{(W_{gate} \cdot 10^6)^{\mathbf{NSUBBWP}}} \right) \quad (134)$$

10.2 Mobility Modification due to Mechanical Stress

It is known that the trench isolation induces mechanical stress in the channel, which results in a degradation of the mobility [17], and causes a reduction of the saturation current with reduced W_{gate} . This effect is modeled by a decreasing phonon mobility with two model parameters **MUEPHW** and **MUEPWP** as

$$M_{uephonon} = M_{uephonon} \cdot \left(1 + \frac{\mathbf{MUEPHW}}{(W_{gate} \cdot 10^6)^{\mathbf{MUEPWP}}} \right) \quad (135)$$

$$M_{uephononb} = M_{uephononb} \cdot \left(1 + \frac{\mathbf{MUEPHWB}}{(W_{gate} \cdot 10^6)^{\mathbf{MUEPWPB}}} \right) \quad (136)$$

However, the I_{ds} - W_{gate} characteristic does not show monotonous decrease but starts to increase for narrower W_{gate} . This effect is modeled as a change of the surface-roughness contribution caused by a carrier flow in increasing distance from the surface as

$$M_{uesurface} = M_{uesurface} \cdot \left(1 + \frac{\mathbf{MUESRW}}{(W_{gate} \cdot 10^6)^{\mathbf{MUESWP}}} \right) \quad (137)$$

$$M_{uesurfaceb} = M_{uesurfaceb} \cdot \left(1 + \frac{\mathbf{MUESRWB}}{(W_{gate} \cdot 10^6)^{\mathbf{MUESWPB}}} \right) \quad (138)$$

$$V_{max} = V_{max} \cdot \left(1 + \frac{\mathbf{VOVERW}}{wl^{\mathbf{VOVERWP}}} \right) \quad (139)$$

10.3 Transistor Leakage due to Shallow Trench Isolation (STI): Hump in I_{ds}

The shallow trench isolation induces also an undesired hump in the subthreshold region of the I_{ds} - V_{gs} characteristics. A MOSFET leakage current occurs at the STI edges, which is smaller than the main MOSFET current, and only important for modeling of the subthreshold characteristics of the MOSFET. The surface potential of the leakage regions at the trench edges can be derived analytically as in [18]

$$\phi_{S,STI} = V'_{gs,STI} + \frac{\epsilon_{Si} Q_{N,STI}}{C_{FOX}^{'2}} \left[1 - \sqrt{1 + \frac{2C_{FOX}^{'2}}{\epsilon_{Si} Q_{N,STI}} \left(V'_{gs,STI} - V_{bs} - \frac{1}{\beta} \right)} \right] \quad (140)$$

where

$$Q_{N,STI} = q \cdot N_{STI} \quad (141)$$

$$V'_{gs,STI} = V_{gs} - \mathbf{VFBC} + V_{thSTI} + \Delta V_{th,SCSTI} \quad (142)$$

where

$$V_{thSTI} = \mathbf{VTHSTI} - \mathbf{VDSTI} \cdot V_{ds} \quad (143)$$

and

$$\Delta V_{th,SCSTI} = \frac{\epsilon_{Si}}{C_{FOX}} W_{d,STI} \frac{dE_y}{dy} \quad (144)$$

The threshold voltage for the STI effect **VTHSTI** includes features of STI such as N_{STI} , which are different from the substrate. The depletion-layer thickness $W_{d,STI}$ is written as

$$W_{d,STI} = \sqrt{\frac{2\epsilon_{Si}(2\Phi_{B,STI} - V_{bs})}{q \cdot N_{STI}}} \quad (145)$$

$\frac{dE_y}{dy}$ is described with model parameters in the same form as in section 4.1 on short-channel effects

$$\frac{dE_y}{dy} = \frac{2(\mathbf{VBI} - 2\Phi_{B,STI})}{(L_{gate,sm} - \mathbf{PARL2})^2} (\mathbf{SCSTI1} + \mathbf{SCSTI2} \cdot V_{ds}), \quad (146)$$

where

$$L_{gate,sm} = L_{gate} + \frac{\mathbf{WL1}}{wl} \mathbf{WL1P} \quad (147)$$

$$wl = (W_{gate} \cdot 10^6) \cdot (L_{gate} \cdot 10^6) \quad (148)$$

The modeling of the transistor leakage for STI technologies is based on the idea that the current in the subthreshold region is governed only by the diffusion term. The carrier concentration $Q_{i,STI}$ is calculated analytically by the Poisson equation with the substrate-impurity concentration of N_{STI} different from **NSOI** and **NSUBP**. The final leakage current equation is written as

$$I_{ds,STI} = 2 \frac{W_{STI}}{L_{eff} - \Delta L} \mu \frac{Q_{i,STI}}{\beta} [1 - \exp(-\beta V_{ds})] \quad (149)$$

where W_{STI} determines the width of the high-field region.

The gate length/width dependence of N_{STI} and W_{STI} are considered as

$$N_{STI} = \mathbf{NSTI} \left(1 + \frac{\mathbf{NSTIL}}{(L_{gate,sm} \cdot 10^6) \mathbf{NSTILP}} \right) \left(1 + \frac{\mathbf{NSTIW}}{(N_{gate,sm} \cdot 10^6) \mathbf{NSTIWP}} \right) \quad (150)$$

$$W_{STI} = \mathbf{WSTI} \left(1 + \frac{\mathbf{WSTIL}}{(L_{gate,sm} \cdot 10^6) \mathbf{WSTILP}} \right) \left(1 + \frac{\mathbf{WSTIW}}{(W_{gate,sm} \cdot 10^6) \mathbf{WSTIWP}} \right) \quad (151)$$

where

$$W_{gate,sm} = W_{gate} + \frac{\mathbf{WL1}}{wl} \mathbf{WL1P} \quad (152)$$

To consider the non-planar surface due to the STI technology, **RATWSTI** is introduced to modify the intrinsic width.

$$W_{STI,mod} = W_{STI} \cdot \mathbf{RATWSTI} \quad (153)$$

To consider the current caused by the STI effect, the flag **COISTI** must be set to 1, where the default is set to zero.

10.4 Small Geometry

Small size devices do not show the same scaling characteristic as long-channel or wide-channel devices, but rather deviate significantly. The small geometry effects are modeled first as a threshold voltage shift which is portion of the threshold voltage shift (equation (49), Section 4).

$$\Delta V_{th,sm} = \frac{\mathbf{WL2}}{wl \mathbf{WL2P}} \quad (154)$$

$$wl = (W_{gate} \cdot 10^6) \cdot (L_{gate} \cdot 10^6) \quad (155)$$

where W_{gate} and L_{gate} are in meter (m). With the use of the scaling factor 10^6 (m^{-1}), the quantities $W_{gate} \cdot 10^6$ and $L_{gate} \cdot 10^6$ become dimensionless.

The mobility modification due to the small device geometry is also modeled in the phonon scattering as

$$M_{uephonon} = M_{uephonon} \cdot \left(1 + \frac{\mathbf{MUEPHS}}{wl \mathbf{MUEPSP}} \right) \quad (156)$$

$$M_{uephononb} = M_{uephononb} \cdot \left(1 + \frac{\mathbf{MUEPHSB}}{wl \mathbf{MUEPSPB}} \right) \quad (157)$$

$$V_{max} = V_{max} \cdot \left(1 + \frac{\mathbf{VOVERS}}{wl \mathbf{VOVERSP}} \right) \quad (158)$$

Table 9: HiSIM_SOTB model parameters introduced in Section 10 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

front * MUEPHW * MUEPWP * MUESRW * MUESWP * MUEPHS * MUEPSP	phonon-related mobility reduction phonon-related mobility reduction surface roughness-related mobility change surface roughness-related mobility change mobility change due to small size mobility change due to small size
back * MUEPHWB * MUEPWPB * MUESRWB * MUESWPB * MUEPHSB * MUEPSPB	phonon-related mobility reduction phonon-related mobility reduction surface roughness-related mobility change surface roughness-related mobility change mobility change due to small size mobility change due to small size
WFC * WVTH0 * NSUBP0 * NSUBWP NSUBSW NSUBSWP NSUBSMAX NSUBBW NSUBBWP * VOVERW * VOVERWP	threshold voltage change due to cap change threshold voltage shift modification of pocket concentration for narrow W modification of pocket concentration for narrow W W dependence of NSUBS W dependence of NSUBS upper limit of SOI layer concentration W dependence of NSUBB W dependence of NSUBB W dependence of velocity overshoot W dependence of velocity overshoot
STI * VTHSTI * VDSTI * SCSTI1 * SCSTI2 NSTI * NSTIL * NSTILP * NSTIW * NSTIWP WSTI * WSTIL * WSTILP * WSTIW * WSTIWP * RATWSTI WL1 WL1P	threshold voltage shift due to STI V_{ds} dependence of STI subthreshold the same effect as SC1 but at STI edge the same effect as SC2 but at STI edge substrate-impurity concentration at the STI edge channel-length dependence of NSTI channel-length dependence of NSTI channel-width dependence of NSTI channel-width dependence of NSTI width of the high-field region at STI edge channel-length dependence of WSTI channel-length dependence of WSTI channel-width dependence of WSTI channel-width dependence of WSTI ratio of WSTI threshold voltage shift of STI leakage due to small size effect threshold voltage shift of STI leakage due to small size effect
WL2 WL2P * VOVERS * VOVERSP	threshold voltage shift due to small-size effect threshold voltage shift due to small-size effect modification of max velocity due to small size modification of max velocity due to small size

11 Well-Proximity Effect

Impurity concentration of the channel region within the well (**NSUBS**, **NSUBP**) is dependent on the distance between the well edge and the gate edge. The channel length is also a factor to modify the impurity concentration. This becomes obvious as the distance reduces. To characterize the layout features, three specific factors are determined by CMC; **SCA**, **SCB**, **SCC** [19]. The use of these factors the well-proximity effect is model as

$$\mathbf{NSUBS} = \mathbf{NSUBS} + \mathbf{NSUBSWPE} \cdot (\mathbf{SCA} + \mathbf{WEB} \cdot \mathbf{SCB} + \mathbf{WEC} \cdot \mathbf{SCC}) \quad (159)$$

$$\mathbf{NSUBP} = \mathbf{NSUBP} + \mathbf{NSUBPWPE} \cdot (\mathbf{SCA} + \mathbf{WEB} \cdot \mathbf{SCB} + \mathbf{WEC} \cdot \mathbf{SCC}) \quad (160)$$

where **SCA**, **SCB**, and **SCC** are instance parameters determined from the layout, and **NSUBSWPE**, **NSUBPWPE**, **WEB**, and **WEC** are model parameters.

The HiSIM_SOTB model parameters introduced in section 11 are summarized in Table 10.

Table 10: HiSIM_SOTB model parameters introduced in section 11 of this manual. # indicates instance parameters.

# SCA	layout characterization factor
# SCB	layout characterization factor
# SCC	layout characterization factor
NSUBSWPE	channel concentration change due to WPE
NSUBPWPE	pocket concentration change due to WPE
WEB	modification of layout characterization factor
WEC	modification of layout characterization factor

12 Effects of the Source/Drain Diffusion Length for Shallow Trench Isolation (STI) Technologies

The diffusion length, L_{od} between MOSFET gate and STI edge affects the MOSFET characteristics.

The influence is observed mainly in V_{th} and in the saturation current, namely in the mobility. The V_{th} change is attributed to a change of the impurity concentration and modeled as

$$N_{substi} = \frac{1 + T1 \cdot T2}{1 + T1 \cdot T3} \quad (161)$$

where

$$T1 = \frac{1}{1 + \mathbf{NSUBSSTI2}} \quad (162)$$

$$T2 = \left(\frac{\mathbf{NSUBSSTI1}}{L_{od, half}} \right)^{\mathbf{NSUBSSTI3}} \quad (163)$$

$$T3 = \left(\frac{\mathbf{NSUBSSTI1}}{L_{od, half, ref}} \right)^{\mathbf{NSUBSSTI3}} \quad (164)$$

which is used to modify the channel concentration N_{subs} as

$$N_{subs} = N_{subs} \cdot N_{substi}. \quad (165)$$

The V_{th} change is attributed to a change of the pocket impurity concentration and modeled as

$$T_{substi} = \frac{1 + T1 \cdot T2}{1 + T1 \cdot T3} \quad (166)$$

where

$$T1 = \frac{1}{1 + \mathbf{NSUBPSTI2}} \quad (167)$$

$$T2 = \left(\frac{\mathbf{NSUBPSTI1}}{L_{od, half}} \right)^{\mathbf{NSUBPSTI3}} \quad (168)$$

$$T3 = \left(\frac{\mathbf{NSUBPSTI1}}{L_{od, half, ref}} \right)^{\mathbf{NSUBPSTI3}} \quad (169)$$

which is used to modify the pocket concentration N_{subp} as

$$N_{subps} = \begin{cases} N_{subpp} \cdot T_{substi} & (L_{od, half} > 0) \\ N_{subpp} & (L_{od, half} \leq 0) \end{cases} \quad (170)$$

The saturation-current change is attributed to a change of the mobility and modeled as

$$M_{uesti} = \frac{1 + T1 \cdot T2}{1 + T1 \cdot T3} \quad (171)$$

where

$$T1 = \frac{1}{1 + \mathbf{MUESTI2}} \quad (172)$$

$$T2 = \left(\frac{\mathbf{MUESTI1}}{L_{od, half}} \right)^{\mathbf{MUESTI3}}$$

$$T3 = \left(\frac{\mathbf{MUESTI1}}{L_{od, half, eff}} \right)^{\mathbf{MUESTI3}} \quad (172)$$

which is used to modify the phonon mobility parameter $M_{uephonon}$ as

$$M_{uephonon} = M_{uephonon} \cdot M_{uesti} \quad (173)$$

where $L_{od, half}$ and $L_{od, half, eff}$ are determined in the same way as BSIM4.6.0 [23] with model parameters **SAREF** and **SBREF** and instance parameters **SA**, **SB**, and **SD**.

Table 11: HiSIM_SOTB model parameters introduced in Section 12 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

NSUBSSTI1	channel concentration change due to diffusion-region length between gate and STI
NSUBSSTI2	channel concentration change due to diffusion-region length between gate and STI
NSUBSSTI3	channel concentration change due to diffusion-region length between gate and STI
NSUBPSTI1	pocket concentration modifier
NSUBPSTI2	pocket concentration modifier
NSUBPSIT3	pocket concentration modifier
MUESTI1	mobility change due to diffusion length
MUESTI2	mobility change due to diffusion length
MUESTI3	mobility change due to diffusion length
SAREF	ref-dist between OD edge to poly of one side
SBREF	ref-dist between OD edge to poly of the other side
SA	length of diffusion between gate and STI
SB	length of diffusion between gate and STI
SD	length of diffusion between gate and gate

13 Temperature Dependence

If **TEMP** is treated as a model parameter, temperature T is determined as

$$T = \mathbf{TEMP} + \mathbf{DTEMP} \quad (174)$$

where **DTEMP** is the temperature increase by self-heating. **DTEMP** can also be an instance parameter **DTEMP**. If **TEMP** is treated as an instance parameter, T is determined as

$$T = \mathbf{TEMP} \quad (175)$$

The temperature dependence is included automatically in the surface potentials through β , which is the inverse of the thermal voltage. Additionally the bandgap, the intrinsic carrier concentration, the carrier mobility, and the carrier saturation velocity are also temperature dependent. The temperature dependence of the bandgap [20] determines the temperature dependence of V_{th} [20] and is modeled as

$$E_g = E_{g,TNOM} - \mathbf{BGTMP1} \cdot (T - \mathbf{TNOM}) - \mathbf{BGTMP2} \cdot (T^2 - \mathbf{TNOM}^2) \quad (176)$$

where T is the given temperature, and

$$E_{g,TNOM} = \mathbf{EG0} - 90.25 \times 10^{-6} \cdot \mathbf{TNOM} - 1.0 \times 10^{-7} \cdot \mathbf{TNOM}^2 \quad (177)$$

The temperature dependence of the intrinsic carrier concentration is given by

$$n_i = n_{i0} \cdot T^{\frac{3}{2}} \cdot \exp\left(-\frac{E_g}{2q}\beta\right) \quad (178)$$

The temperature dependence of the mobility and the temperature dependence of the saturation velocity have a major influence on the temperature dependence of the I_{ds} - V_{ds} characteristics under the on-current condition. They are modeled as [15]:

$$\mu_{PH}(\text{phonon}) = \frac{M_{uephonon}}{(T/\mathbf{TNOM})^{M_{tmp}} \times E_{eff}^{MUEPH0}} \quad (179)$$

$$M_{tmp} = \mathbf{MUETMP} \cdot \left(1 + \frac{\mathbf{MUETMPL}}{(L_{gate} \cdot 10^6)^{MUETMPLP}}\right) + \mathbf{MUETMP1} \cdot (T/\mathbf{TNOM})^2 \quad (180)$$

$$V_{max}(T) = \frac{V_{max}}{1.8 + 0.4 \cdot (T/\mathbf{TNOM}) + 0.1 \cdot (T/\mathbf{TNOM})^2 - V_{tmp} \cdot (1 - T/\mathbf{TNOM})} \quad (181)$$

$$V_{tmp} = \mathbf{VTMP} \cdot \left(1 + \frac{\mathbf{VTMPL}}{(L_{gate} \cdot 10^6)^{VTMPLP}}\right) \quad (182)$$

$$I_{GIDL}(T) = \frac{I_{GIDL}}{\left(1.0 - \exp\left(\frac{-L_{eff}}{\mathbf{GIDLBPL1} \cdot (T/\mathbf{TNOM})^{\mathbf{GIDLBPLT}}}\right)\right)} \quad (183)$$

In addition to the temperature dependence of the physical quantities considered, resistances include the temperature dependence, which is modeled with the given temperature to avoid complication in parameter extraction.

$$\mu_{drift0,temp} = \frac{\mathbf{RDRMUED}}{(T/\mathbf{TNOM})^{\mathbf{RDRMUETMP}}} \quad (184)$$

$$\mu_{source0,temp} = \frac{\mathbf{RDRMUES}}{(T/\mathbf{TNOM})^{\mathbf{RDRMUETMP}}} \quad (185)$$

$$(186)$$

$$V_{max_drift,temp} = \frac{\mathbf{RDRVMAXD}}{1.8 + 0.4(T/\mathbf{TNOM}) + 0.1(T/\mathbf{TNOM})^2 - \mathbf{RDRV TMP} \cdot (1 - T/\mathbf{TNOM})} \quad (187)$$

$$V_{max_source,temp} = \frac{\mathbf{RDRVMAXS}}{1.8 + 0.4(T/\mathbf{TNOM}) + 0.1(T/\mathbf{TNOM})^2 - \mathbf{RDRV TMP} \cdot (1 - T/\mathbf{TNOM})} \quad (188)$$

$$R_{drbb,temp} = \mathbf{RDRBBD} + \mathbf{RDRBB TMP}(T - \mathbf{TNOM}) \quad (189)$$

$$R_{srbb,temp} = \mathbf{RDRBBS} + \mathbf{RDRBB TMP}(T - \mathbf{TNOM}) \quad (190)$$

Table 12: HiSIM_SOTB model parameters introduced in Section 13 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

TEMP	device temperature
DTEMP	device temperature change
BGTMP1	temperature dependence of bandgap
BGTMP2	temperature dependence of bandgap
TNOM	temperature selected as a nominal value
EG0	bandgap
MUETMP	temperature-dependence of phonon scattering
MUETMPL	temperature-dependence of phonon scattering
MUETMPLP	temperature-dependence of phonon scattering
MUETMP1	temperature-dependence of phonon scattering
*VTMP	temperature-dependence coefficient
VTMPL	temperature-dependence coefficient
VTMPLP	temperature-dependence coefficient
VOTMP	temperature-dependence coefficient
VOTMP2	temperature-dependence coefficient
GIDLBPL1	length for GIDL/GISL parasitic bipolar effect
GIDLBPLT	temperature effect for GIDLBPL1
RDRBBD	high field mobility in drift region (drain side)
RDRBBS	high field mobility in drift region (source side)
RDRMUED	mobility in drift region (drain side)
RDRMUES	mobility in drift region (source side)
RDRVMAXD	saturation velocity in drift region (drain side)
RDRVMAXS	saturation velocity in drift region (source side)
RDRVTMP	temperature dependence of resistance
RDRMUETMP	temperature dependence of resistance
RDRBBTMP	temperature dependence of resistance

14 Resistances

Figure 7 shows a schematic feature of the device. The potential drops V_{ssp} and V_{ddp} are calculated by SPICE iteratively.

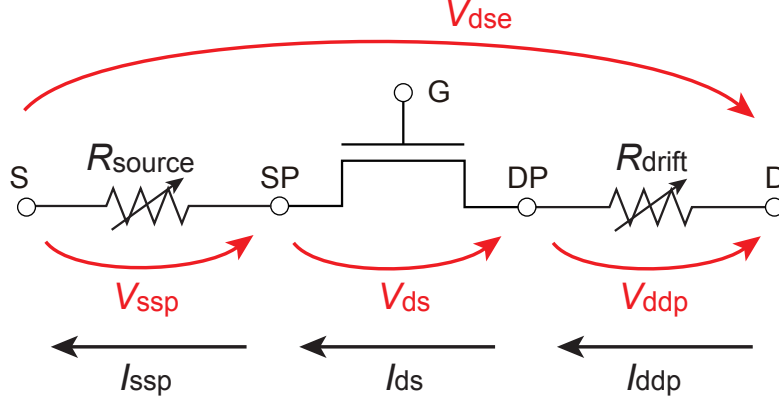


Fig. 7: Model concept

14.1 Drain-side resistance

The drain side resistance R_{drift} is written with the current flowing in the drift region I_{ddp} as

$$R_{drift} = \left(\frac{V_{ddp}}{I_{ddp}} + \mathbf{RSH} \cdot \mathbf{NRD} \right) \quad (191)$$

where $\mathbf{NRD}$ is an instance parameter describing the number of squares of the drain diffusion, and $\mathbf{RSH}$ is its the sheet resistance of the square. The first terms of the right hand side of Eq. (191) considers the resistance in the drift region, and the second term is the that in the diffusion region, which are layout dependent.

$$I_{ddp} = W_{eff} \cdot \mathbf{NF} \cdot X_{ov} \cdot q \cdot \mathbf{NOVER} \cdot \mu_{drift} \frac{V_{ddp}}{\mathbf{LDRIFT}} \quad (192)$$

where

$$\mu_{drift} = \frac{\mu_{drift0}}{\left[1 + \left(\frac{\mu_{drift0}}{V_{max,drift}} \cdot \frac{V_{ddp}}{\mathbf{LDRIFT}} \right)^{R_{drbb,temp}} \right]^{\frac{1}{R_{drbb,temp}}}} \quad (193)$$

$$\mu_{drift0} = \mu_{drift0,temp} \left(1 + \frac{\mathbf{RDRMUEL}}{(L_{gate} \cdot 10^6)^{\mathbf{RDRMUPL}}} \right) \quad (194)$$

$$V_{max,drift} = V_{max,drift,temp} \left(1 + \frac{\mathbf{RDRVMAXL}}{(L_{gate} \cdot 10^6)^{\mathbf{RDRVMAXLP}}} \right) \left(1 + \frac{\mathbf{RDRVMAXW}}{(W_{gate} \cdot 10^6)^{\mathbf{RDRVMAXWP}}} \right) \quad (195)$$

$$X_{ov} = \sqrt{\mathbf{XLD}^2 + \mathbf{RDRDJUNC}^2} \quad (196)$$

The potential value of V_{ddp} , the difference between the internal node DP and the drain node is calculated by circuit simulator.

14.2 Source-side resistance

The source-side resistance R_{source} is written with the current flowing in the drift region I_{ssp} as

$$R_{\text{source}} = \left(\frac{V_{\text{ssp}}}{I_{\text{ssp}}} + \mathbf{RSH} \cdot \mathbf{NRS} \right) \quad (197)$$

where $\mathbf{NRS}$ is an instance parameter describing the number of squares of the source diffusion, and $\mathbf{RSH}$ is its sheet resistance of the square. The first terms of the right hand side of Eq. (197) considers the resistance in the source drift region, and the second term is the that in the source diffusion region, which is layout dependent.

$$I_{\text{ssp}} = W_{\text{eff}} \cdot \mathbf{NF} \cdot X_{\text{ov}} \cdot q \cdot \mathbf{NOVERS} \cdot \mu_{\text{source}} \frac{V_{\text{ssp}}}{\mathbf{LDRIFTS}} \quad (198)$$

where

$$\mu_{\text{source}} = \frac{\mu_{\text{source0}}}{\left[1 + \left(\frac{\mu_{\text{source0}}}{V_{\text{max_source}}} \cdot \frac{V_{\text{ssp}}}{\mathbf{LDRIFTS}} \right)^{R_{\text{srbb,temp}}} \right]^{\frac{1}{R_{\text{srbb,temp}}}}} \quad (199)$$

$$\mu_{\text{source0}} = \mu_{\text{source0,temp}} \left(1 + \frac{\mathbf{RDRMUEL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{RDRMUPL}}} \right) \quad (200)$$

$$V_{\text{max_source}} = V_{\text{max_source,temp}} \left(1 + \frac{\mathbf{RDRVMAXL}}{(L_{\text{gate}} \cdot 10^6)^{\mathbf{RDRVMAXLP}}} \right) \left(1 + \frac{\mathbf{RDRVMAXW}}{(W_{\text{gate}} \cdot 10^6)^{\mathbf{RDRVMAXWP}}} \right) \quad (201)$$

$$X_{\text{ov}} = \sqrt{\mathbf{XLD}^2 + \mathbf{RDRDJUNC}^2} \quad (202)$$

The potential value of V_{ssp} , the difference between the internal node SP and the drain node is calculated by circuit simulator.

14.3 Gate resistance

The gate resistance becomes large as the gate width becomes large, which is the case for RF circuits. The equation for the gate-resistance calculation is taken from BSIM4 [23] description as

$$R_g = \frac{\mathbf{RSHG} \cdot \left(\mathbf{XGW} + \frac{W_{\text{eff}}}{3 \cdot \mathbf{NGCON}} \right)}{\mathbf{NGCON} \cdot (L_{\text{drawn}} - \mathbf{XGL}) \cdot \mathbf{NF}}, \quad (203)$$

where $\mathbf{RSHG}$ is the gate sheet resistance, and others are instance parameters dependent on layout. The flag $\mathbf{CORG}$ is provided for the inclusion of gate resistance. $\mathbf{CORG} = 0, 1$ means “no”, “external” gate resistance, respectively.

The HiSIM_SOTB model parameters introduced in section 14 are summarized in Table 13.

Table 13: HiSIM_SOTB model parameters introduced in Section 14 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

RSHG	gate sheet resistance
XGW	distance from gate contact to channel edge
NGCON	number of gate contacts
XGL	offset of gate length
RSH	sheet resistance of diffusion region (drain side)
#NRS	number of source squares
#NRD	number of drain squares
#LDRIFT	length of drift region (drain side)
#LDRIFTS	length of drift region (source side)
NOVERS	impurity concentration in overlap region (source side)
RDRDJUNC	junction depth at channel/drift region (drain side)
RDRVMAXL	saturation velocity L_{gate} dependence
RDRVMAXLP	saturation velocity L_{gate} dependence
RDRVMAXW	saturation velocity W_{gate} dependence
RDRVMAXWP	saturation velocity W_{gate} dependence
RDRMUEL	mobility in drift region L_{gate} dependence
RDRMUELP	mobility in drift region L_{gate} dependence

15 Capacitances

15.1 Intrinsic Capacitances

HiSIM_SOTB uses analytical solutions for all independent intrinsic capacitances, derived from the charges as explicit functions of the surface potentials. Therefore, there are no new model parameters for the intrinsic capacitances except the channel width modification parameter **XWDC** (see in Section 1).

The lateral electric field along the channel induces a capacitance C_{Q_y} which significantly affects the gate capacitance in saturation [21]. The induced charge associated with C_{Q_y} is described with the surface potential values as

$$Q_y = \epsilon_{Si} W_{effc} \cdot \mathbf{NF} \cdot W_d \left(\frac{\phi_{S0} + V_{ds} - \phi_S(\Delta L)}{\mathbf{XQY}} \right) + \frac{\mathbf{XQY1} \cdot W_{effc} \cdot 10^6 \cdot \mathbf{NF}}{(L_{gate} \cdot 10^6) \mathbf{XQY2}} V_{es} \quad (204)$$

introducing **XQY**, a parameter determining the maximum field at the channel/drain junction independent of L_{gate} . For **XQY**=0 the charge Q_y is fixed to zero. To compensate the enhanced short-channel effect, determined by the current characteristics, two model parameters **XQY1** and **XQY2** are introduced. To activate Q_y the model parameter **CLM1** must be smaller than unity.

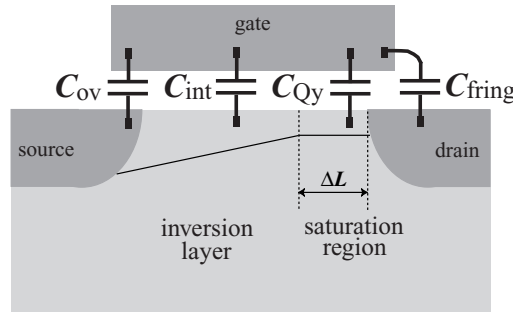


Fig. 8: Modeling of the gate-drain capacitance with C_{Q_y} added to the conventional components.

15.2 Overlap Capacitances

The overlap charge at the drain side Q_{god} is written as

$$\frac{Q_{god}}{W_{effc} \cdot \mathbf{NF} \cdot C_{FOX}} = \int_0^{\mathbf{LOVER}} (V_{gs} - \phi_s) dy \quad (205)$$

Similarly, at the source side,

$$\frac{Q_{gos}}{W_{effc} \cdot \mathbf{NF} \cdot C_{FOX}} = \int_0^{\mathbf{LOVERS}} (V_{gs} - \phi_s) dy \quad (206)$$

Thus the surface-potential distribution along the gate overlap region determines the charge and the capacitance.

To consider the overlap capacitance the flag **COADOV** must be selected to 1, which is the default. The overlap capacitance includes three options as summarized in Fig. 9, where the drain side and the source side are treated as identical. If Flag **COOVLP**=0, the overlap capacitances are treated to be constant. If **CGSO** and **CGDO** are provided, these values are taken. If they are not provided, the values are calculated with the overlap length and oxide capacitance.

The default overlap capacitance flag (**COOVLP** = 0) calculates bias-independent drain and source overlap capacitances from C_{FOX} as

$$C_{gso} = C_{FOX} \cdot \mathbf{LOVER} \cdot W_{effc} \quad (207)$$

$$C_{gdo} = C_{FOX} \cdot \mathbf{LOVER} \cdot W_{effc} \quad (208)$$

User-defined values can be specified using the input parameters **CGDO** and **CGSO**. If these values are not specified, the overlap capacitances are calculated using

$$C_{ov} = -\frac{\epsilon_{ox}}{T_{FOX}} \mathbf{LOVER} \cdot W_{effc} \cdot \mathbf{NF} \quad (209)$$

The gate-to-bulk overlap capacitance $C_{gbo.loc}$ is calculated only with a user-defined value **CGBO** using

$$C_{gbo.loc} = -\mathbf{CGBO} \cdot L_{gate} \quad (210)$$

independent of the model flag **COOVLP**.

If Flags **COOVLP**=1, the bias dependent overlap capacitances are considered. Here two models are provided: One is the surface-potential-based model and the other describes with a simple V_{gs} dependence. In addition to the bias dependent capacitances, **CGSO** and **CGDO** can be also added, if they are provided.

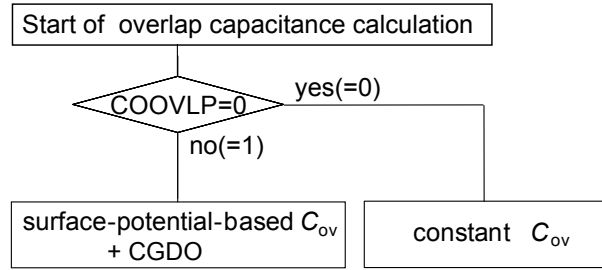


Fig. 9: Model options of the overlap capacitance at the drain side are summarized.

For the source side, the same calculation is performed with $V_{ds}=0$.

The surface potential ϕ_s is calculated in the same manner as in the channel region, and only the polarity is inverted from the channel. The final overlap charge equation is written with the calculated ϕ_s

$$Q_{over} = W_{effc} \cdot \mathbf{NF} \cdot \mathbf{LOVER} \cdot C_{FOX} (V_{gs} - V_{bs} - \mathbf{VFBOVER} - \phi_s) \quad (211)$$

where **LOVER** is the length of the overlap region of the gate over drain, **NOVER** is the impurity concentration in the drift region, and **VFBOVER** is the flat-band voltage in the overlap region. This model is selected, if **NOVER** is not equal to zero.

Two options are provided to calculate ϕ_s , which is selected by the flag **COQOVSM**:

COQOVSM=0: with an analytical equation excluding inversion charge

COQOVSM=1: with iterative procedure (default)

15.3 Extrinsic Capacitances

The outer fringing capacitance is modeled as [22]

$$C_{fringe} = \frac{\epsilon_{ox}}{\pi/2} W_{gate} \cdot \mathbf{NF} \cdot \ln \left(1 + \frac{\mathbf{TPOLY}}{T_{FOX}} \right) \quad (212)$$

where **TPOLY** is the gate-poly thickness. This capacitance is bias independent.

Another extrinsic capacitance is the one between the gate protruded over substrate which accounts for extra capacitance. The model for this capacitance model is active only when a body-tie device is considered.

Table 14: HiSIM_SOTB model parameters introduced in Section 15 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

XQY	distance from junction to max field point
XQY1	V_{es} -dependence of Q_y
XQY2	L_{gate} -dependence of Q_y
CGSO	gate-to-source overlap cap
CGDO	gate-to-drain overlap cap
CGBO	gate-to-bulk overlap cap
LOVER	overlap length
NOVER	impurity concentration in overlap region
VFBOVER	flat-band voltage in overlap region
TPOLY	height of poly- S_i gate for fringing cap

16 Parasitic Currents

16.1 Body Current

Impact ionization in the SOI layer causes an additional current, though it is not measured as substrate current on SOTB-MOSFETs but as parasitics on top of the intrinsic drain current. As significant portion of present formulation in this section inherits from that for a conventional substrate current model, so do many variables and parameters here.

After some simplification we derive the well-known equation [24]

$$I_{\text{sub}} = \frac{C_1}{C_2} (\phi(y) - \phi(0)) I_{\text{ds,SUB}} \exp\left(-\frac{\lambda C_2}{\phi(y) - \phi(0)}\right) \quad (213)$$

where

$$\lambda^2 = \frac{\epsilon_{\text{Si}} X_j T_{\text{FOX}}}{\epsilon_{\text{ox}}} \quad (214)$$

and X_j is the junction depth and $I_{\text{ds,SUB}}$ is the simplified source-drain current which is determined analytically as (cf. Section 17)

$$I_{\text{ds,SUB}} = \frac{W_{\text{eff}}}{L_{\text{eff}}} \mu_{\text{sub}} \frac{Q_{\text{i,SUB}}}{\beta} [1 - \exp(-\beta V_{\text{ds}})] \quad (215)$$

where μ_{sub} is 300, and $Q_{\text{i,SUB}}$ is calculated in the same way as shown in Eq. (20) with the analytical surface potential as

$$\phi_{\text{STI}} = \phi_{\text{sb,STI}} - 0.5 \cdot \left(\phi_{\text{sab,STI}} + \sqrt{\phi_{\text{sab,STI}}^2 + 4 \cdot \mathbf{SUBDLT} \cdot \phi_{\text{sb,STI}}} \right) \quad (216)$$

$$\phi_{\text{sab,STI}} = \phi_{\text{sb,STI}} - \phi_{\text{sa,STI}} - \mathbf{SUBDLT} \quad (217)$$

$$\phi_{\text{sb,STI}} = \frac{\ln(A_{\text{STI}} \cdot V'_{\text{gp,SUB}})}{\beta + \frac{2}{V'_{\text{gp,SUB}}}} \quad (218)$$

$$A_{\text{STI}} = \frac{\beta \cdot C_{\text{FOX}}^2 \cdot N_{\text{subs}}^2}{2 \cdot q \cdot \epsilon_{\text{Si}} \cdot n_i^2 \cdot N_{\text{subs}}} \quad (219)$$

$$\phi_{\text{sa,STI}} = V'_{\text{gp,SUB}} + \frac{\epsilon_{\text{Si}} q N_{\text{subs}}}{C_{\text{FOX}}^2} \left[1 - \sqrt{1 + \frac{2C_{\text{FOX}}'^2}{\epsilon_{\text{Si}} q N_{\text{subs}}} \left(V'_{\text{gp,SUB}} - \frac{1}{\beta} \right)} \right] \quad (220)$$

$$V'_{\text{gp,SUB}} = V_{\text{gs}} - V_{\text{FBSUBI}} - \Delta V_{\text{th}} - \phi_{\text{Spg}} \quad (221)$$

where V_{FBSUBI} is the L_{gate} -dependent flat-band voltage for I_{sub} .

$$V_{\text{FBSUBI}} = \mathbf{VFBSUB} \left(1 + \frac{\mathbf{VFBSUBL}}{L_{\text{gate}} \mathbf{VFBSUBLP}} \right) \quad (222)$$

The basic equation of Eq. (213) is modified so that measurements for all device sizes can be reproduced.

$$I_{\text{sub}} = X_{\text{sub1}} \cdot P_{\text{sisubsat}} \cdot I_{\text{ds,SUB}} \cdot \exp\left(-\frac{X_{\text{sub2}}}{P_{\text{sisubsat}}}\right) \quad (223)$$

where

$$X_{\text{sub1}} = \text{SUB1} \cdot \left(1 + \frac{\text{SUB1L}}{L_{\text{gate}}^{\text{SUB1LP}}} \right) \quad (224)$$

$$X_{\text{sub2}} = \text{SUB2} \cdot \left(1 + \frac{\text{SUB2L}}{L_{\text{gate}}} \right) \quad (225)$$

$$P_{\text{sisubsat}} = \text{SVDS} \cdot V_{\text{ds}} + \phi_{\text{STI}} - \frac{L_{\text{gate}}}{L_{\text{gate}} + \text{SLG}} \left(1 + \frac{\text{SVGSW}}{W_{\text{gate}}^{\text{SVGSWP}}} \right) P_{\text{sislsat}} \quad (226)$$

$$P_{\text{sislsat}} = V'_{\text{gp,SUB}} \cdot \text{SVGSI} + \frac{q \cdot \epsilon_{\text{Si}} \cdot N_{\text{subs}}}{C_{\text{FOX}}^2} \cdot \left\{ 1 - \sqrt{\left(1 + \frac{2C_{\text{FOX}}^2}{q \cdot \epsilon_{\text{Si}} \cdot N_{\text{subs}}} \right) \cdot \left(V'_{\text{gp,SUB}} - \frac{1}{\beta} - X_{\text{vbs}} \cdot V_{\text{es}} \right)} \right\} \quad (227)$$

$$\text{SVGSI} = \text{SVGS} \left(1 + \frac{\text{SVGSL}}{L_{\text{gate}}^{\text{SVGSLP}}} \right) \quad (228)$$

$$X_{\text{vbs}} = \text{SVBS} \cdot \left(1 + \frac{\text{SVBSL}}{L_{\text{gate}}^{\text{SVBSLP}}} \right) \quad (229)$$

Impact-Ionization Induced Body Potential Change

Impact ionization generates electron and hole pairs, which is the origin of the body current. However, not only the parasitic current but also the charge distribution in the SOI layer is changed. This induced charge redistribution affects as the body potential change. This is modeled in a simple way as

$$I_{\text{ds,BPC}} = \frac{2}{3} \sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subs}}}{\beta}} \left[\left\{ \beta(\phi_{\text{sL}} - V_{\text{es}}) - 1 \right\}^{\frac{3}{2}} \frac{3}{2} \frac{\beta \Delta V_{\text{body}}}{\beta(\phi_{\text{sL}} - V_{\text{es}}) - 1} - \left\{ \beta(\phi_{\text{s0}} - V_{\text{es}}) - 1 \right\}^{\frac{3}{2}} \frac{3}{2} \frac{\beta \Delta V_{\text{body}}}{\beta(\phi_{\text{s0}} - V_{\text{es}}) - 1} \right] - \sqrt{\frac{2\epsilon_{\text{Si}}qN_{\text{subs}}}{\beta}} \left[\left\{ \beta(\phi_{\text{sL}} - V_{\text{es}}) - 1 \right\}^{\frac{1}{2}} \frac{1}{2} \frac{\beta \Delta V_{\text{body}}}{\beta(\phi_{\text{sL}} - V_{\text{es}}) - 1} - \left\{ \beta(\phi_{\text{s0}} - V_{\text{es}}) - 1 \right\}^{\frac{1}{2}} \frac{1}{2} \frac{\beta \Delta V_{\text{body}}}{\beta(\phi_{\text{s0}} - V_{\text{es}}) - 1} \right] \quad (230)$$

where

$$\Delta V_{\text{body}} = \text{IBPC1} \cdot (1 + \text{IBPC2} \cdot \Delta V_{\text{th}}) \cdot I_{\text{sub}} \quad (231)$$

IBPC1 and **IBPC2** are model parameters.

16.2 Front Gate Tunneling Current

(i) Between Gate and Channel, I_{gate}

As for the current between gate and channel, (I_{gate}) the direct-tunneling mechanism is considered [25]. Since measured I_{gate} shows nearly linear L_{gate} dependence, the tunneling is assumed to occur along the whole channel length. Thus the final description implemented in HiSIM_SOTB is [26, 27]

$$I_{\text{gate}} = q \cdot \text{GLEAK1} \cdot \frac{E^2}{E_{\text{gp}}^{\frac{1}{2}}} \cdot \exp \left(-\frac{E_{\text{gp}}^{\frac{3}{2}} \cdot \text{GLEAK2}}{E^2} \right) \cdot \left(\frac{Q_{\text{i}}}{\text{const0}} \right)^{\text{GLEAK9}} \cdot W_{\text{eff}} \cdot \text{NF} \cdot L_{\text{eff}} \cdot \frac{\text{GLEAK6}}{\text{GLEAK6} + V_{\text{ds}}} \cdot \frac{\text{GLEAK7}}{\text{GLEAK7} + W_{\text{eff}} \cdot \text{NF} \cdot L_{\text{eff}}} \quad (232)$$

where

$$E = \left(1 + \frac{E_y}{\mathbf{GLEAK5}}\right) \cdot \left(1 - \frac{1}{1 + V_{gs}^2}\right) \cdot \frac{V_G}{T_{FOX}} \quad (233)$$

$$V_G = V_{gs} - \mathbf{GLEAK8} \cdot V_{fb} + \frac{(\mathbf{GLEAK4} \cdot \Delta V_{th} - \mathbf{GLEAK10} \cdot V_{es})}{L_{eff}} - \mathbf{GLEAK3} \cdot \phi_s(\Delta L) \quad (234)$$

ΔV_{th} is threshold voltage shift introduced in Section 4 (Threshold Voltage Shift), $\phi_s(\Delta L)$ is surface potential at the channel/drain junction and was introduced in Section 9 (Channel-Length Modulation). const0 is the same constant introduced in Section 2.

GLEAK1 through **GLEAK8** are model parameters, and E_{gp} describes the temperature dependent bandgap for the gate current (see Section 13. The gate-channel current I_{gate} is partitioned into two terminal currents with one model parameter in the following manner:

$$I_{gate} = I_{gate,s} + I_{gate,d} \quad (235)$$

where

$$I_{gate,s} = (1 - P_{partition}) \cdot I_{gate} \quad (236)$$

$$I_{gate,d} = P_{partition} \cdot I_{gate} \quad (237)$$

where an analytical description of $P_{partition}$ is obtained by integrating the following equation.

$$P_{partition} = \frac{1}{I_{gate}} \int_0^{L_{eff}} \frac{y}{L_{eff}} I_{gate}(y) dy \quad (238)$$

(ii) Between Gate and Bulk, I_{gb}

The I_{gb} current under the accumulation condition is modeled as

$$I_{gb1} = \mathbf{GLKB1} \cdot E_{gb}^{\mathbf{GLKB5}} \cdot \exp\left(-\frac{\mathbf{GLKB2}}{E_{gb}^{\mathbf{GLKB6}}}\right) W_{eff} \cdot \mathbf{NF} \cdot f1(LG) \quad (239)$$

$$E_{gb} = -\frac{V_{gs} - \mathbf{GLKB4} \cdot V_{es} - V_{fb} - \mathbf{GLKB3}}{T_{FOX}} \quad (240)$$

where $f1$ is an auxiliary function expressed as

$$f1(LG) = L_{gate} \cdot 10^6 + \mathbf{GLKB7} \quad (241)$$

and its value is clamped so that $f1(LG) > \mathbf{GLKB8}$.

$$I_{gb2} = \mathbf{GLKB21} \cdot E_{gb}^{\mathbf{GLKB25}} \cdot \exp\left(-\frac{\mathbf{GLKB22}}{E_{gb}^{\mathbf{GLKB26}}}\right) W_{eff} \cdot \mathbf{NF} \cdot f2(LG) \quad (242)$$

$$E_{gb} = -\frac{V_{gs} - \mathbf{GLKB24} \cdot V_{es} - V_{fb} - \mathbf{GLKB23}}{T_{FOX}} \quad (243)$$

where $f2$ is an auxiliary function expressed as

$$f2(LG) = L_{gate} \cdot 10^6 + \mathbf{GLKB27} \quad (244)$$

where its value is clamped so that $f2(LG) > \mathbf{GLKB28}$.

(iii) Between Gate and Source/Drain, I_{gs}/I_{gd}

The tunneling current between the gate and the source/drain overlap region is modeled as

$$I_{gs} = \text{sign} \cdot \mathbf{GLKSD1} \cdot E_{gs}^2 \exp(T_{FOX}(-\mathbf{GLKSD2} \cdot V_{gs} + \mathbf{GLKSD3})) W_{eff} \cdot \mathbf{NF} \cdot f1(LG) \quad (245)$$

$$f1(LG) = (L_{gate} \cdot 10^6)^{\mathbf{GLKSD4}} \quad (246)$$

$$E_{gs} = \frac{\mathbf{GLKSD5} \cdot V_{gs}}{T_{FOX}} \quad (247)$$

$$I_{gd} = \text{sign} \cdot \mathbf{GLKSD1} \cdot E_{gd}^2 \exp(T_{FOX}(\mathbf{GLKSD2} \cdot (-V_{gs} + V_{ds}) + \mathbf{GLKSD3})) W_{\text{eff}} \cdot \mathbf{NF} \cdot f2(LG) \quad (248)$$

$$f2(LG) = (L_{\text{gate}} \cdot 10^6)^{\mathbf{GLKSD4}} \quad (249)$$

$$E_{gd} = \frac{\mathbf{GLKSD5} \cdot (V_{gs} - V_{ds})}{T_{FOX}} \quad (250)$$

$$\text{sign} = +1 \quad \text{for} \quad E < 0$$

$$\text{sign} = -1 \quad \text{for} \quad E \geq 0$$

16.3 GIDL (gate-induced drain leakage)

The GIDL current is generated at the drain junction under the accumulation condition. The V_{ds} increase induces a very narrow potential well in the drain just under the gate, causing carrier generation. Therefore, the GIDL current is strongly dependent on V_{ds} . At further reduced V_{gs} values the direct gate tunneling starts to dominate the I_{GIDL} measurements, resulting in V_{ds} independence. The V_{ds} dependent I_{GIDL} is modeled here. The generation mechanism is considered to be the direct tunneling between the above mentioned narrow potential well of length ΔY and the ordinary drain region. The generation occurs only in this ΔY region at the drain. The final equation is

$$I_{\text{GIDL}} = q \cdot \mathbf{GIDL1} \cdot \frac{E^2}{E_g^{\frac{1}{2}}} \cdot \exp\left(-\mathbf{GIDL2} \cdot \frac{E_g^{\frac{3}{2}}}{E}\right) \cdot W_{\text{eff}} \cdot \mathbf{NF} \cdot T_1 \quad (251)$$

where

$$T_1 = \frac{1}{(1.0 + \exp(-\beta \cdot V_{ds}))} \quad (252)$$

$$E = \frac{\mathbf{GIDL3} \cdot (V_{ds} + \mathbf{GIDL4}) - V'_G}{T_{FOX}\mathbf{GIDL}} \quad (253)$$

and

$$V'_G = V_{gs} + (\Delta V_{\text{th,SC}} + \Delta V_{\text{th,P}}) \cdot \mathbf{GIDL5} \quad (254)$$

where $\Delta V_{\text{th,SC}}$ and $\Delta V_{\text{th,P}}$ are defined in Section 4. It is recommended to leave $\mathbf{GIDL4}$ at its default value, zero, because I_{GIDL} does not approach zero even under the condition of $V_{gs} = V_{ds} = 0$.

The GISL current is calculated with the same equation as the GIDL current described above. The selection either I_{GIDL} or I_{GISL} is done by the polarity of the current flow.

16.4 Valence band electron tunneling current

Electrons in the channel can reach to the gate electrode through oxide from the valence band as the applied gate voltage gets higher than the band gap. In HiSIM_SOTB, this tunneling current is called as I_{evb} (Electron Valence Band tunneling current) and modeled as

$$I_{\text{evb}} = \mathbf{EVB1} \cdot q \cdot \left(\frac{\phi_b}{V_{FOX}}\right) \left(\frac{2\phi_b}{V_{FOX}} - 1\right) E_{FOX}^2 \cdot \exp\left[-\frac{\mathbf{EVB2} \left(1 - \left(1 - \frac{V_{FOX}}{\phi_b}\right)^{\frac{3}{2}}\right)}{E_{FOX}}\right] \cdot W_{\text{eff}} \cdot \mathbf{NF} \cdot L_{\text{eff}} \quad (255)$$

where ϕ_b is the barrier height for holes and is fixed to 4.12 eV. Here E_{FOX} and V_{FOX} are described as

$$E_{FOX} = -\frac{\mathbf{FVBS} \cdot V_{es} - V_{FOX} + \Delta V_{\text{th,SC}} + \Delta V_{\text{th,P}} + E_g + \mathbf{EVB3}}{T_{FOX}} \quad (256)$$

where

$$V_{FOX} = V'_G - \phi_s \quad (257)$$

and E_g is the silicon bandgap.

Table 15: HiSIM_SOTB model parameters introduced in Section 16 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

VFBSUB	flatband voltage for I_{sub} calculation
VFBSUBL	L_{gate} -dependence of VFBSUB
VFBSUBLP	L_{gate} -dependence of VFBSUB
SUB1	substrate current coefficient of magnitude
SUB1L	L_{gate} -dependence of SUB1
SUB1LP	L_{gate} -dependence of SUB1
SUB2	substrate current coefficient of exponential term
SUB2L	L_{gate} -dependence of SUB2
SUBDLT	smoothing parameter
SVDS	substrate current dependence on V_{ds}
SLG	substrate current dependence on L_{gate}
SVGSW	W_{gate} -dependence of SVGS
SVGSWP	W_{gate} -dependence of SVGS
SVGS	substrate current dependence on V_{gs}
SVGSL	L_{gate} -dependence of SVGS
SVGSLP	L_{gate} -dependence of SVGS
SVBS	substrate current dependence on V_{es}
SVBSL	L_{gate} -dependence of SVBS
SVBSLP	L_{gate} -dependence of SVBS
IBPC1	impact-ionization-induced bulk potential change
IBPC2	impact-ionization-induced bulk potential change
Gate leakage current	
GLEAK1	gate-to-channel current coefficient
GLEAK2	gate-to-channel current coefficient
GLEAK3	gate-to-channel current coefficient
GLEAK4	gate-to-channel current coefficient
GLEAK5	G-t-C short channel correction
GLEAK6	G-t-C V_{ds} -dependence correction
GLEAK7	G-t-C L and W dependence correction
GLEAK8	gate-to-channel current coefficient
GLEAK9	gate-to-channel current coefficient
GLKB1	G-t-S/D current coefficient
GLKB2	G-t-S/D current coefficient
GLKB3	G-t-S/D current coefficient
GLKB4	G-t-S/D current coefficient
GLKB5	G-t-S/D current coefficient
GLKB6	G-t-S/D current coefficient
GLKB7	G-t-S/D current coefficient
GLKB8	G-t-S/D current coefficient
GLKB21	G-t-S/D current coefficient
GLKB22	G-t-S/D current coefficient
GLKB23	G-t-S/D current coefficient
GLKB24	G-t-S/D current coefficient
GLKB25	G-t-S/D current coefficient
GLKB26	G-t-S/D current coefficient
GLKB27	G-t-S/D current coefficient
GLKB28	G-t-S/D current coefficient
GLKSD1	G-t-S/D current coefficient
GLKSD2	G-t-S/D current coefficient
GLKSD3	G-t-S/D current coefficient
GLKSD4	G-t-S/D current coefficient
GLKSD5	G-t-S/D current coefficient

Gate-induced drain/source leakage	
GIDL1	magnitude of GIDL
GIDL2	field-dependence of GIDL
GIDL3	V_{ds} -dependence of GIDL
*GIDL4	threshold for V_{ds} dependence
*GIDL5	high-field correction
EVB1	electron tunneling from valence band
EVB2	electron tunneling from valence band
EVB3	electron tunneling from valence band
FVBS	V_{es} dependence of Fowler-Nordheim current

17 Floating-Body Effect Model

The hole storage in the SOI layer Q_h is one of the most important phenomena which causes the floating-body effect (FBE).

Model selector **COFBE** as well as **COISUB** must be set to 1 to activate FBE model. In HiSIM_SOTB, Q_h is modeled as follows:

$$Q_h = \left\{ [\exp[-\beta(\phi_{s0} - \Delta V_{sb})] + \beta(\phi_{s0} - \Delta V_{sb}) - 1]^{1/2} - [\exp[-\beta\phi_{s0}] + \beta\phi_{s0} - 1]^{1/2} \right\} \quad (258)$$

$$\Delta V_{sb} = \mathbf{QHE1} \cdot \frac{1}{\beta} \log \left[1 + \frac{(I_{sub} + I_{evb})L_p L_n}{q \cdot \mathbf{TSOI} \cdot W_{eff} e^{-\beta \cdot \mathbf{QHE2}} (D_n N_d L_p + D_p N_d L_n)} \right] \quad (259)$$

where

$$L_n = \sqrt{D_n \cdot 10^{-7}} \quad (260)$$

$$L_p = \sqrt{D_p \cdot 10^{-7}} \quad (261)$$

$$N_d = 10^{20} cm^{-3} \quad (262)$$

$$D_n = 36 cm^2/sec \quad (263)$$

$$D_p = 13 cm^2/sec \quad (264)$$

Resulting Q_h is included in the Poisson equation (see Eq. (8)) for the consistent inclusion of the FBE. HiSIM_SOTB solves Eqs. (8), (11), and (12) to find surface potentials.

The impact-ionization current and FBE calculation steps in HiSIM_SOTB are summarized below with the flowchart in Fig. 10.

Table 18: HiSIM_SOTB model parameters introduced in Section 17 of this manual.

QHE1	FBE parameter
QHE2	FBE parameter

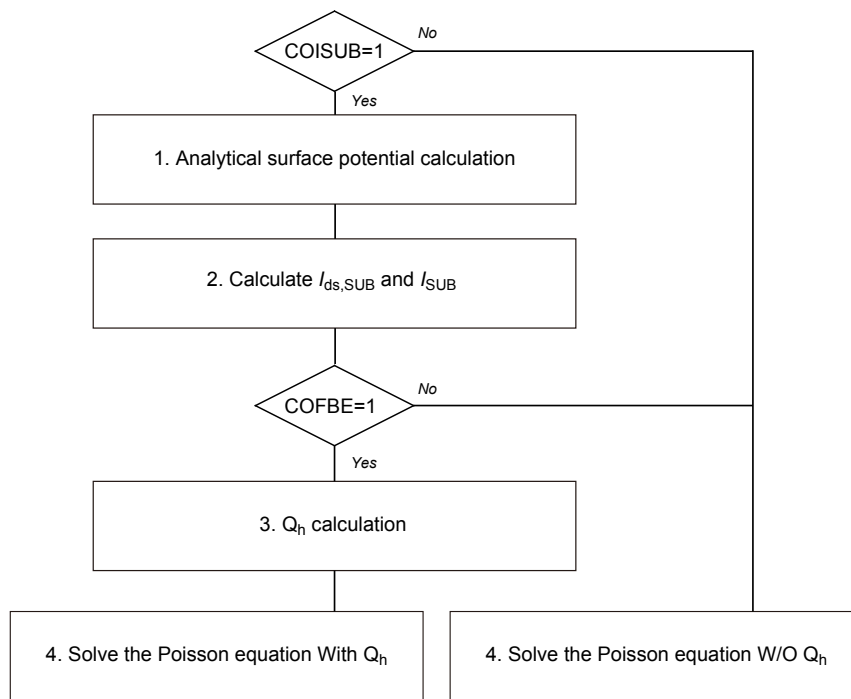


Fig. 10: Calculation flow of the impact-ionization current and floating-body effect in HiSIM_SOTB

18 History Effect Model

Time constant for the history effect is modelled as

$$\tau_h = R_{sb} \cdot C_{FOX} \quad (265)$$

$$R_{sb} = \frac{\mathbf{HIST1}}{I_{sub} + \mathbf{HIST2}} \quad (266)$$

where **HIST1** and **HIST2** are model parameters.

History effect is described with the same formula as the NQS effect as

$$Q_h(t) = Q_h(t - \Delta t) + \frac{\Delta t}{\tau_h + \Delta t} \cdot (Q_{h0} - Q_h(t - \Delta t)) \quad (267)$$

where Q_{h0} is the quasi-static hole storage in the SOI layer without charging delay shown in Eq. (258) in Section 17.

Table 19: HiSIM_SOTB model parameters introduced in Section 18 of this manual. * indicates parameters that can be used for this portion of the model.

HIST1	history-effect parameter
HIST2	history-effect parameter

19 Self-Heating Effect Model

The self-heating effect is modeled as a C-R thermal network (Fig. 11) with a heat resistance parameter **RTH0** and capacitance parameter **CTH0**. The flag **COSELFHEAT** must be equal to 1 and **RTH0** must not be equal to zero to activate this model. The SHE should be switched on/off only with the model flag **COSELFHEAT**.

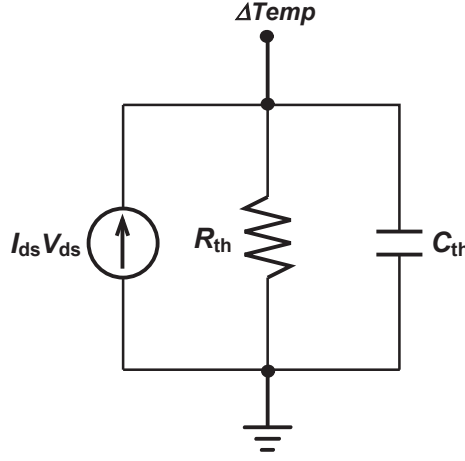


Fig. 11: Thermal Network applied for the self-heating effect. $\text{Temp} \leftarrow \text{Temp} + \Delta\text{Temp}$. ΔTemp is the incremental temperature change derived from the self-heating effect in each time step and Temp is the accumulated temperature change in comparison to the initial temperature.

Table 20: HiSIM_SOTB model parameters introduced in Section 19 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

RTH0	thermal resistance
CTH0	thermal capacitance

20 Noise Models

20.1 $1/f$ Noise Model

The $1/f$ noise is caused by both the carrier fluctuation and the mobility fluctuation [29]

$$S_{I_{ds}} = \frac{I_{ds}^2 \mathbf{NFTRP}}{\beta f (L_{\text{eff}} - \Delta L) W_{\text{eff}} \cdot \mathbf{NF}} \left[\frac{1}{(N_0 + N^*)(N_L + N^*)} + \frac{2\mu_f E_y \mathbf{NFALP}}{N_L - N_0} \ln \left(\frac{N_L + N^*}{N_0 + N^*} \right) + (\mu_f E_y \mathbf{NFALP})^2 \right] \quad (268)$$

where the parameters **NFALP** and **NFTRP** represent the contribution of the mobility fluctuation and the ratio of trap density to attenuation coefficient, respectively. N_0 and N_L are carrier densities at source side and drain side or pinch-off point, respectively, as calculated in HiSIM_SOTB. N^* is written as

$$N^* = \frac{C_{\text{FOX}} + C_{\text{dep}} + \mathbf{CIT}}{q\beta} \quad (269)$$

where C_{dep} is the depletion capacitance calculated with ϕ_s . **CIT** is the capacitance caused by the interface-trapped carriers and is normally fixed to be zero.

$$N_{\text{flick}} = S_{I_{ds}} \cdot f \quad (270)$$

is calculated in HiSIM_SOTB.

20.2 Thermal Noise Model

The equations for S_{id} in our compact-modeling approach, become functions of the self-consistent surface potentials as well as the surface-potential derivatives at source and drain [30, 31].

$$S_{id} = 4kT \frac{W_{eff} \cdot \mathbf{NF} \cdot C_{FOX} V_{gvt} \mu_f}{(L_{eff} - \Delta L)} \cdot \frac{(1 + 3\eta + 6\eta^2)\mu_d^2 + (3 + 4\eta + 3\eta^2)\mu_d\mu_f + (6 + 3\eta + \eta^2)\mu_f}{15(1 + \eta)\mu_{av}^2} \quad (271)$$

with

$$\mu_d = \left(1.0 + \left(\frac{\mu_0 E_{yd}}{V_{max,therm}} \right)^{BB} \right)^{\frac{1}{BB}} \quad (272)$$

$$\mu_{av} = \frac{\mu_f + \mu_d}{2.0} \quad (273)$$

where μ_f , μ_d , and μ_{av} are mobilities at the source side (see Section 8), the drain side, averaged, respectively. E_{yd} is derived from calculated ϕ_s values. $V_{max,therm}$ is maximum electron-saturation velocity (1.0×10^7 cm/s). V_{gvt} denotes an effective gate voltage that is equal to the charge density at the source side Q_{i0} divided by the oxide capacitance C_{FOX} . const0 is the same constant introduced in Section 2.

The quantity η in (271) is calculated as

$$\eta = 1 - \frac{(\phi_{sL} - \phi_{s0}) + \chi(\phi_{sL} - \phi_{s0})}{V_{gvt}} \quad (274)$$

$$\text{where } \chi = 2 \frac{\text{const0}}{C_{FOX}} \left[\left[\frac{2}{3} \frac{1}{\beta} \frac{\{\beta(\phi_{sL} - V_{es}) - 1\}^{\frac{3}{2}} - \{\beta(\phi_{s0} - V_{es}) - 1\}^{\frac{3}{2}}}{\phi_{sL} - \phi_{s0}} \right] - \sqrt{\beta(\phi_{s0} - V_{es}) - 1} \right] \quad (275)$$

Thus no additional model parameters are required for the thermal noise model.

$$N_{thrm1} = S_{id}/4kT \quad (276)$$

is calculated in HiSIM_SOTB.

20.3 Induced Gate Noise Model

No additional model parameters are required for the induced gate noise model.

$$N_{igate} = S_{igate}/f^2 \quad (277)$$

is calculated in HiSIM_SOTB. Explicit model equation is presented in [32].

20.4 Coupling Noise Model

No additional model parameters are required for the coupling noise model.

$$N_{cross} = \frac{S_{igid}}{\sqrt{S_{igate} \cdot S_{id}}} \quad (278)$$

is calculated in HiSIM_SOTB. Explicit model equation is in [32].

Table 21: HiSIM_SOTB model parameters introduced in Section 20 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

NFTRP	ratio of trap density to attenuation coefficient
NFALP	contribution of mobility fluctuation
*CIT	cap caused by the interface trapped carriers

21 Non-Quasi-Static (NQS) Model

This section briefly describes the NQS model incorporated in HiSIM_SOTB. For the full equations, please see reference [34].

21.1 Carrier Formation

Carriers in the channel take time to build-up as opposed to the Quasi-Static (QS) approximation. To consider this phenomenon in HiSIM_SOTB, the carrier formation is modeled as [33, 34]

$$q(t_i) = \frac{q(t_{i-1}) + \frac{\Delta t}{\tau} Q(t_i)}{1 + \frac{\Delta t}{\tau}} \quad (279)$$

where $q(t_i)$ and $Q(t_i)$ represent the non-quasi-static and the quasi-static carrier density at time t_i , respectively, and $\Delta t = t_i - t_{i-1}$ is valid. Eq. (279) implies that the formation of carriers under the NQS approximation is always delayed in comparison to the QS approximation, which is the basic origin of the NQS effect. The delay is determined by the carrier transit delay τ and the time interval in the circuit simulation Δt . The transit delay for Q_i is τ (equation (282)) and that for Q_b is τ_b (equation (283)).

21.2 Delay Mechanisms

Two charges, one is the inversion charge and the other is the bulk charge, are considered. The transit delay of Q_i is τ and that of Q_b is τ_b .

Up to weak inversion, carriers diffuse to the channel and the transit delay is approximated by

$$\tau_{\text{diff}} = \mathbf{DLY1} \quad (280)$$

At strong inversion, there is already conduction due to field-driven carriers. The transit delay due to the flow of such conductive carriers is

$$\tau_{\text{cond}} = \mathbf{DLY2} \cdot \frac{Q_i}{I_{\text{ds}}} \quad (281)$$

where **DLY2** is a constant coefficient. These two delay mechanisms (diffusion and conduction) are combined using the Matthiessen rule

$$\frac{1}{\tau} = \frac{1}{\tau_{\text{diff}}} + \frac{1}{\tau_{\text{cond}}} \quad (282)$$

Applying the same approach for the formation of body carriers, leads to the approximation of the body carrier delay as an RC delay in the form

$$\tau_b = \mathbf{DLY3} \cdot C_{\text{FOX}} \quad (283)$$

where **DLY3** is a constant coefficient and C_{FOX} is the oxide capacitance.

21.3 Time-Domain Analysis

The total drain/source/body terminal currents are derived from the superposition of the transport current and the charging current. The transport current is a function of the instantaneous terminal voltages and is approximated by the steady-state solution. The source/drain/body charging currents are the time derivatives of the associated non-quasi-static charges, q_s , q_D , and q_B , respectively.

21.4 AC Analysis

The non-quasi-static charge formulation in the time-domain is transformed into the frequency-domain using the Fourier transformation. The resulting NQS charge $\hat{q}$ is derived as [35]

$$\hat{q}_a(\omega) = \left\{ \frac{1}{1 + (\tau\omega)^2} - j \frac{\tau\omega}{1 + (\tau\omega)^2} \right\} \hat{Q}_a(\omega) \quad (284)$$

where $\hat{Q}$ is the QS charge in the frequency-domain and τ is the carrier transit delay. The NQS capacitances are calculated as

$$C_{ab} \equiv \frac{\partial q_a}{\partial V_b} = -\frac{2(\tau\omega)^2}{(1 + (\tau\omega)^2)^2} \frac{1}{\tau} \frac{\partial \tau}{\partial V_b} \cdot Q_{a,QS} + \frac{1}{1 + (\tau\omega)^2} \cdot C_{ab,QS} - i \left[\frac{\tau\omega(1 - (\tau\omega)^2)}{(1 + (\tau\omega)^2)^2} \frac{1}{\tau} \frac{\partial \tau}{\partial V_b} \cdot Q_{a,QS} + \frac{\tau\omega}{1 + (\tau\omega)^2} \cdot C_{ab,QS} \right] \quad (285)$$

where $C_{ab,QS}$ is the quasi-static capacitance. a and b denote the MOSFET terminals; gate g , drain d , source s and bulk b . For y -parameter calculation, conventional circuit simulators implement the quasi-static description as given for example in [16]. In order to obtain the NQS y -parameters, the capacitances in the conventional equations are replaced by the NQS capacitances derived in Eq. (285).

In general, y_{ab} is obtained as

$$y_{ab} = \frac{i\omega}{1 + (\tau\omega)^2} [C_{ab,QS} + (\tau\omega)^2 A_{ab}(\omega) - i[\tau\omega B_{ab}(\omega) + \tau\omega C_{ab,QS}]] \quad (286)$$

where A_{ab} and B_{ab} are defined as

$$A_{ab} = \frac{1}{\tau} \frac{-2}{1 + (\tau\omega)^2} \frac{\partial \tau}{\partial V_b} \cdot Q_{a,QS} \quad (287)$$

$$B_{ab} = \frac{1}{\tau} \frac{1 - (\tau\omega)^2}{1 + (\tau\omega)^2} \frac{\partial \tau}{\partial V_b} \cdot Q_{a,QS} \quad (288)$$

y_{gg} , for example, is explicitly is written as

$$y_{gg} = \frac{i\omega}{1 + (\tau\omega)^2} [C_{gg,QS} + (\tau\omega)^2 A_{gg}(\omega) - i[\tau\omega B_{gg}(\omega) + \tau\omega C_{gg,QS}]] \quad (289)$$

Due to the described approach, the non-quasi static AC-analysis model implemented in HiSIM_SOTB requires no additional model parameters.

Table 22: HiSIM_SOTB model parameters introduced in Section 21 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

DLY1	coefficient for delay due to diffusion of carriers
DLY2	coefficient for delay due to conduction of carriers
DLY3	coefficient for RC delay of bulk carriers

22 Multiplication Factor

For the case of a multiple device construction the multiple factor M is introduced as an instance parameter. All device features such as currents and capacitances are multiplied by M . Noises are also multiplied by M .

Table 23: HiSIM_SOTB model parameters introduced in Section 22 of this manual. * indicates parameters that can be used to fine-tune this portion of the model.

M	multiplication factor
-----	-----------------------

23 Conservation of Symmetry at $V_{ds} = 0$

HiSIM_SOTB preserves the symmetry at $V_{ds} = 0$ automatically due to the drift-diffusion approximation. However, modeling of the short-channel effects induces a small asymmetry. To eliminate the asymmetry caused by the artifacts of the modeling, the V_{th} modeling has to include a damping of the short-channel effects as V_{ds} approaches zero. This corresponds to a vanishing of short-channel effects as V_{ds} gets near to zero, which is actually also observed in 2D simulations. In HiSIM_SOTB the damping is done by a mathematical function with two parameters: **VZADD0** and **PZADD0**. The values of these parameters are fixed, and it is recommended not to change them.

Table 24: HiSIM_SOTB model parameters introduced in Section 23 of this manual.

VZADD0	symmetry conservation coefficient
PZADD0	symmetry conservation coefficient

24 Flags for Model Selection

24.1 Model selection flags

Following flags are prepared to select required model options.

1. Overlap capacitance model is selected as:
COOVLP = 0: constant overlap capacitance (default)
COOVLP = 1: bias dependent overlap capacitance based on surface potential calculation
2. Overlap capacitance model selector (continued):
COQOVSM = 0: analytical evaluation only (iterative solution for surface potential not to be invoked)
COQOVSM = 1: iteratively calculates surface potential values for evaluating overlap capacitances
3. Substrate current I_{sub} is calculated:
COISUB = 0: no (default)
COISUB = 1: yes
4. Gate current I_{gate} is calculated:
COIGS = 0: no (default)
COIGS = 1: yes
5. GIDL current I_{GIDL} is calculated:
COGIDL = 0: no (default)
COGIDL = 1: yes
6. STI leakage current $I_{\text{ds,STI}}$ is calculated:
COISTI = 0: no (default)
COISTI = 1: yes
7. Lateral field induced and overlap charges/capacitances are added to intrinsic ones:
COADOV = 0: no
COADOV = 1: yes (default)
8. Non-quasi-static mode is invoked:
CONQS = 0: no (default)
CONQS = 1: yes
9. Source resistance is included:
CORS = 0: no (default)
CORS = 1: yes
10. Drain resistance is included:
CORD = 0: no (default)
CORD = 1: yes
11. Gate-contact resistance is included (This flag can also be given as a instance parameter):
CORG = 0: no (default)
CORG = 1: yes
12. $1/f$ noise is calculated:
COFLICK = 0: no (default)
COFLICK = 1: yes
13. Thermal noise is calculated:
COTHRML = 0: no (default)
COTHRML = 1: yes
14. Induced gate and cross correlation noise are calculated:
COIGN = 0 || **COTHRML** = 0: no (default)
COIGN = 1 & **COTHRML** = 1: yes

15. The Poisson equation is solved iteratively, where the previous time-step value ϕ_S is used for the initial value:
COPPRV = 0: no
COPPRV = 1: yes (default)
16. Self-Heating:
COSELFHEAT = 0: no (default)
COSELFHEAT = 1: yes
17. Floating-Body Effect:
COFBE = 0: no (default)
COFBE = 1: yes (COISUB=1 is also required)
18. History Effect:
COHIST = 0: no (default)
COHIST = 1: yes
19. Valence-band-electron-tunneling-current model:
COIEVB = 0: no (default)
COIEVB = 1: yes
20. Symmetry treatment:
COVBSBIZ = 0: no (default)
COVBSBIZ = 1: yes
21. Inversion capacitance flag:
COCINV = 0: no (default)
COCINV = 1: yes
22. New mobility model flag:
CONEWMUB = 0: no (default)
CONEWMUB = 1: yes

24.2 Disabling models by setting model parameters

To exclude specific modeled effects, the following parameter settings should be chosen.

Short-channel effect	SC1 = SC2 = SC3 = 0
Reverse-short-channel effect	LP = 0
Quantum-mechanical effect	QME1 = QME3 = 0
Channel-length modulation	CLM1 = CLM2 = CLM3 = 0
Narrow-channel effect	WFC = MUEPHW = 0
Small-size effect	WL2 = 0

25 List of model parameters, default values, and ranges

The maximum and minimum limits of the model parameter are recommended values. * indicates parameters that can be used to fine-tune this portion of the model.

Parameter	Unit	Min	Default	Max	Description	See
Basic device parameters						
TYPE	-	-1	1	1	1 for nMOS and -1 for pMOS	
TFOX	m	$0 <$	3.5 n		front oxide thickness	Fig. 1
TBOX	m	$0 <$	10 n		buried oxide thickness	Fig. 1
TSOI	m	$0 <$	10 n		silicon film thickness	Fig. 1
XLD	m	$0 \leq$	0	≤ 50 n	gate overlap length	(3)
XLDL	m		$20.0 \cdot 10^{-9}$		channel-length dependence of XLD	
XLDLMIN	m		$10.0 \cdot 10^{-9}$		minimum value of XLDL	
XWD	m	-10 n $\leq$	0	≤ 50 n	gate overlap width	(4)
XWDC	m		0		different width dependence of cap. from currents	(5)
TPOLY	m	$0 \leq$	0		height of poly-Si gate for fringing cap	(212)
NSUBS	cm^{-3}	$1 \cdot 10^{15} \leq$	$1 \cdot 10^{17}$	$\leq 1 \cdot 10^{19}$	SOI layer impurity concentration	Fig. 1
NSUBB	cm^{-3}	$1 \cdot 10^{15} \leq$	$1 \cdot 10^{18}$		substrate impurity concentration	Fig. 1
NSUBBL	cm^{-3}		0.0		channel-length dependence of NSUBB	
NSUBBLP	cm^{-3}		1.0		channel-length dependence of NSUBB	
NSUBBW			0.0		W dependence of NSUBB	
NSUBBWP			1.0		W dependence of NSUBB	
NSUBBMIN	cm^{-3}		$1.0 \cdot 10^{14}$		minimum values of NSUBB	
NSUBP	cm^{-3}	$1 \cdot 10^{16} \leq$	$1 \cdot 10^{17}$	$\leq 1 \cdot 10^{19}$	max pocket concentration	(61)
VFBC	V	$-1.2 \leq$	-1	≤ -0.5	flat-band voltage	(14)
VBI	V	$1 \leq$	1.1	≤ 1.2	built-in potential	(51)
VFBC1	V		0.0		channel length dependence of flat-band voltage	
VFBC1P	-		1.0		channel length dependence of flat-band voltage	
VFBC2	V		0.0		channel length dependence of flat-band voltage	
VFBC2P	$\text{cm} \cdot \text{V} \cdot \text{V}^{-1}$		1.0		channel length dependence of flat-band voltage	
VFBHAMP	$\text{V} \cdot \text{cm}^{-1}$		0.0		channel length dependence of flat-band voltage	
VBSND	V		3.0		V_{bs} for smoothing	
VBSMAX	V		3.5		maximum V_{bs}	
smoothing coefficient between linear and saturation region						
DDLTMAX	-	$1 \leq$	10	≤ 20	smoothing coefficient for V_{ds}	(34)
DDLTSPL	μm^{-1}	$0 \leq$	50	≤ 100	L_{gate} -dependence of smoothing coefficient	(36)
DDLTICT	-	$-3 \leq$	0	≤ 20	L_{gate} -dependence of smoothing coefficient	(36)
Mobility and velocity						
MUECB0	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	$100 \leq$	300	≤ 100 k	Coulomb scattering	(78)
MUECB0LP			0.0		length dependence of Coulomb scattering	
MUECB0L2			0.0		length dependence of Coulomb scattering	
MUECB0L2P			1.0		length dependence of Coulomb scattering	
MUECB1	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	$1 \leq$	30	≤ 10 k	Coulomb scattering	(78)
MUECB1LP			0.0		length dependence of Coulomb scattering	
MUECB1L2			0.0		length dependence of Coulomb scattering	
MUECB1L2P			1.0		length dependence of Coulomb scattering	
MUEPH0	-	$0.25 \leq$	0.3	≤ 0.35	phonon scattering	(79)
MUEPH1	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	2 k $\leq$	25 k	≤ 100 k	phonon scattering (default:nMOS=25k, pMOS=9k)	(173)
MUETMP	$\mu\text{m} \cdot \text{MUEPLP}$	$0.5 \leq$	1.5	≤ 2	temperature-dependence of phonon scattering	(179)
MUETMPL			0.0		temperature-dependence of phonon scattering	(179)
MUETMPLP			1.0		temperature-dependence of phonon scattering	(179)
MUETMP1	$\mu\text{m} \cdot \text{MUEPLP}$		0.0		temperature-dependence of phonon scattering	
MUEPHL*	-		0		L-dependence of phonon mobility reduction	(173)
MUEPLP*	-		1		L-dependence of phonon mobility reduction	(173)
MUESR0	-	$1.8 \leq$	2	≤ 2.2	surface-roughness scattering	(96)
MUESR1	$\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$	$1 \cdot 10^{13} \leq$	$2 \cdot 10^{15}$	$\leq 1 \cdot 10^{16}$	surface-roughness scattering	(80)
	$(\text{V cm}^{-1}) \cdot \text{MUESR0}$					
MUESRL*	$\mu\text{m} \cdot \text{MUESLP}$		0		L-dependence of surface roughness on mobility	(96)
MUESLP*	-		1		L-dependence of surface roughness on mobility	(96)
NDEP	-	$0 \leq$	1	≤ 1	depletion charge contribution to E_{eff}	(93)
NINV	-	$0 \leq$	0.5	≤ 1	inversion charge contribution to E_{eff}	(94)
NINVD*	V^{-1}	$0 \leq$	0.0		inversion charge parameter	(83)
NINVDP*	-	$0 \leq$	1.0		inversion charge parameter	(87) (110)
BB	-		2 or 1		high-field-mobility degradation	(99)
VMAX	$\text{cm} \cdot \text{s}^{-1}$	$1 \cdot 10^6 \leq$	$7 \cdot 10^6$	$\leq 2 \cdot 10^7$	(nMOS=2 and pMOS=1)	(101)
VOVER	$\text{cm} \cdot \text{VOVERP}$	$0 \leq$	$10 \cdot 10^{-3}$	≤ 1	saturation velocity	(101)
VOVERP	-	$0 \leq$	$100 \cdot 10^{-3}$	≤ 3	velocity overshoot effect parameter	(101)
VOVERL			0		L dependence of velocity overshoot	
VOVERLP			1		velocity overshoot effect parameter	
VOVERW	?		0		L dependence of velocity overshoot	
VOVERWP	?		1		velocity overshoot effect parameter	
VTMP*	-	$-2 \leq$	0	≤ 1	W dependence of velocity overshoot	(181)
VTMPL	-		0		temperature-dependence of saturation velocity	
VTMPLP	-		1.0		temperature-dependence of saturation velocity	
VOTMP			0.0		temperature-dependence coefficient	
VOTMP2			0.0		temperature-dependence coefficient	
MUEQB			0.0		magnitude of Q_{bL} for front side mobility	
MUEQBL			0.0		length dependence of MUEQB and MUEQBB	

Parameter	Unit	Min	Default	Max	Description	See
MUEQBLP			0.0		length dependence of MUEQB and MUEQBB	
MUEQBB			1.0		magnitude of Q _B L for back side mobility	
NDEPL*			0.0		modify Q _B contribution for short channel	
NDEPLP*			0.0		modify Q _B contribution for short channel	
NINVL			0.0		length dependence of NINV	
NINVLP			0.0		length dependence of NINV	
Mobility Model for back current						
MUECB0B			MUECB0		Coulomb scattering	
MUECB0LPB			MUECB0LP		length dependence of Coulomb scattering	
MUECB0L2B			MUECB0L2		length dependence of Coulomb scattering	
MUECB0L2PB			MUECB0L2P		length dependence of Coulomb scattering	
MUECB1B			MUECB1		Coulomb scattering	
MUECB1LPB			MUECB1LP		length dependence of Coulomb scattering	
MUECB1L2B			MUECB1L2		length dependence of Coulomb scattering	
MUECB1L2PB			MUECB1L2P		length dependence of Coulomb scattering	
MUEPH0B			MUEPH0		phonon scattering	
MUEPH1B			MUEPH1		phonon scattering	
MUEPHLB			MUEPHL		length dependence of phonon mobility	
MUEPLPB			MUEPLP		length dependence of phonon mobility	
MUESR0B			MUESR0		surface-roughness scattering	
MUESR1B			MUESR1		surface-roughness scattering	
MUESRLB			MUESRL		length dependence of surface roughness mobility	
MUESLPB			MUESLP		length dependence of surface roughness mobility	
Short-channel effect						
PARL1	m	0	10 n	≤ 50 n	SOI SCE parameter	(55)
PARL2	m	0	10 n	≤ 50 n	depletion width of channel/contact junction	(51)
SC1	-	0	0	≤ 20	magnitude of short-channel effect	(51)
SC2	V ⁻¹	0	0	≤ 50	V _{ds} -dependence of short-channel effect	(51)
SC3*	m·V ⁻¹	0	0	≤ 1 m	V _{es} dependence of short-channel effect	(51)
SC5	-		0.0		SCE parameter	
SCR1	-	0	0	≤ 5	parameter for SCE via BOX	(55)
SCR2	-	0	0	≤ 5	parameter for SCE via BOX	(55)
SCR3	-	0	0.23	≤ 1	parameter for SCE via BOX	(55)
SCP1	-	0	0	≤ 50	magnitude of short-channel effect due to pocket	(60)
SCP2	V ⁻¹	0	0	≤ 50	V _{ds} -dependence of SCE due to pocket	(60)
SCP3*	m·V ⁻¹	0	0	≤ 1 m	V _{es} -dependence of SCE due to pocket	(60)
LP	m	0	0	≤ 300 n	pocket penetration length	(60)
PTHROU			0.0		subthreshold swing parameter	
VFBSHIFT			0.0		coefficient for V _{fb} shift	
Poly-silicon gate depletion effect						
PGD1	V	0	0	≤ 50 m	strength of poly-depletion effect	(74)
PGD2	V	0	1	≤ 1.5	threshold voltage of poly-depletion effect	(74)
PGD4*	-	0	0	≤ 3	L _{gate} -dependence of poly-depletion effect	(74)
Quantum mechanical effect						
QME1	mV	0	0	≤ 300 n	V _{gs} -dependence of quantum mechanical effect	(76)
QME2	V	0	0	≤ 3	V _{gs} -dependence of quantum mechanical effect	(76)
QME3	m	0	0	≤ 800 p	minimum T _{FOX} modification	(76)
Channel-length modulation						
CLM1	-	0.5	0.7	≤ 1	hardness coefficient of channel/contact junction	(124)
CLM2	-	2	Eq. (127)		coefficient for Q _B contribution	(126)
CLM3	-	1	1	≤ 5	coefficient for Q ₁ contribution	(126)
CLM5	-	0	1	≤ 5	CLM parameter	(128)
CLM6	μm ^{-CLM5}	0	0	≤ 5	CLM parameter	(128)
Punchthrough						
PTL	V ^{PTP-1} m ^{PTLP}	0	0		punchthrough parameter	(71)
PTLP	-		1		punchthrough parameter	(71)
PTP	-	3	3.5	≤ 4	punchthrough parameter	(71)
PT2	V ⁻¹	0	0		punchthrough parameter	(71)
PT4	-	0	0		punchthrough parameter	(71)
PT4P	-	0	1		punchthrough parameter	(71)
*GDL	m ^{GDLP}	0	0	220m	strength of high-field effect	(73)
*GDLP	-		0.0		modification of channel conductance	(73)
*GDLD	m		0		modification of channel conductance	(73)
Narrow-channel effects						
WFC	F·cm ⁻² ·m	-5·10 ⁻¹⁵	0	≤ 1·10 ⁻⁶	Threshold voltage change due to cap change	(130)
WVTH0*	V·μm		0		threshold voltage shift	(129)
NSUBSW**	μm ^{NSUBSWP}		0		W dependence of NSUBS	(133)
NSUBSWP**	-		1		W dependence of NSUBS	(133)
NSUBSMAX**	cm ⁻³		5·10 ¹⁸		upper limit of SOI layer concentration	(133)
NSUBP0*	μm ^{NSUBWP}		0		mod of pocket concentration for narrow W	(134)
NSUBWP*	-		1		mod of pocket concentration for narrow W	(134)
MUEPHW*	μm ^{MUEPHP}		0		phonon-related mobility reduction	(135)
MUEPWP*	-		1		phonon-related mobility reduction	(135)
MUESRW*	μm ^{MUESWP}		0		surface roughness-related mobility change	(137)
MUESWP*	-		1		surface roughness-related mobility change	(137)
WL2	Vμm ^{WL2P}		0		threshold voltage shift due to small-size effect	(154)
WL2P	-		1		threshold voltage shift due to small-size effect	(154)
MUEPHS*	μm ^{MUEPSP}		0		mobility change due to small size	(156)
MUEPSP*	-		1		mobility change due to small size	(156)
VOVERS*	-		0		modification of max velocity due to small size	(158)

Parameter	Unit	Min	Default	Max	Description	See
VOVERSP*	-		1		modification of max velocity due to small size	(158)
Narrow-channel effects for back current						
MUEPHWB	μm^{MUEPHP}		0		phonon-related mobility reduction	(135)
MUEPWPB	-		1		phonon-related mobility reduction	(135)
MUEPHSB	μm^{MUEPSP}		0		mobility change due to small size	(156)
MUEPSPB	-		1		mobility change due to small size	(156)
MUESRWB	μm^{MUESWP}		0		surface roughness-related mobility change	(137)
MUESWPB	-		1		surface roughness-related mobility change	(137)
Well-proximity effect						
NSUBSWPE	$[cm^{-3}]$		0		channel concentration change due to WPE	(159)
NSUBPWPE	$[cm^{-3}]$		0		pocket concentration change due to WPE	(160)
WEB	$[-]$		0		modification of layout characterization factor	(159)
WEC	$[-]$		0		modification of layout characterization factor	(159)
STI effects						
VTHSTI*	V		0		threshold voltage shift due to STI	(143)
VDSTI*	-		0		V_{ds} dependence of STI subthreshold	(143)
SCSTI1*	-		0		the same effect as SC1 but at STI edge	(146)
SCSTI2*	V^{-1}		0		the same effect as SC2 but at STI edge	(146)
NSTI	cm^{-3}	$1 \cdot 10^{16} \leq$	$5 \cdot 10^{17}$	$\leq 1 \cdot 10^{19}$	substrate impurity concentration at STI edge	(141)
NSTIL*	μm^{NSTILP}		0		channel-length dependence of NSTI	(150)
NSTILP*	-		1.0		channel-length dependence of NSTI	(150)
NSTIW*	μm^{NSTIWP}		0		channel-width dependence of NSTI	(150)
NSTIWP*	-		1.0		channel-width dependence of NSTI	(150)
WSTI	m		0		width of high-field region at STI edge	(149)
WSTIL*	μm^{WSTILP}		0		channel-length dependence of WSTI	(151)
WSTILP*	-		1.0		channel-length dependence of WSTI	(151)
WSTIW*	μm^{WSTIWP}		0		channel-width dependence of WSTI	(151)
WSTIWP*	-		1.0		channel-width dependence of WSTI	(151)
RATWSTI*	-		0.0		ratio of WSTI	(153)
WL1	$\mu m^{2WL1P+1}$		0		small-size effect parameter for STI leakage	(147)
WL1P	-		1.0		small-size effect parameter for STI leakage	(147)
NSUBSSTI1	m		0		channel concentration modifier	(163)
NSUBSSTI2	-		0		channel concentration modifier	(162)
NSUBSSTI3	-		1.0		channel concentration modifier	(163)
NSUBPSTI1	m		0		pocket concentration modifier	(169)
NSUBPSTI2	-		0		pocket concentration modifier	(167)
NSUBPSTI3	-		1		pocket concentration modifier	(169)
MUESTI1	m		0		mobility change due to diffusion length	(172)
MUESTI2	-		0		mobility change due to diffusion length	(??)
MUESTI3	-		1		mobility change due to diffusion length	(172)
SAREF	m		$1 \cdot 10^{-6}$		ref-dist between OD edge to poly of one side	p. 29
SBREF	m		$1 \cdot 10^{-6}$		ref-dist between OD edge to poly of the other side	p. 29
Temperature dependence						
EG0	eV	$1 \leq$	1.1785	≤ 1.3	bandgap	(177)
BGTMP1	$eV \cdot K^{-1}$	$50 \mu \leq$	90.25μ	$\leq 100 \mu$	temperature dependence of bandgap	(176)
BGTMP2	$eV \cdot K^{-2}$	$-1 \mu \leq$	0.1μ	$\leq 1 \mu$	temperature dependence of bandgap	(176)
TNOM	$^{\circ}C$	$22 \leq$	27	≤ 32	temperature selected as a nominal value	(176)
Parasitic resistances						
RSHG	Ohm/square		0		gate sheet resistance	(203)
Resistance effect						
LDRIFT	m		1 u		length of drift region (drain side)	(193)
LDRIFTS	m		1 u		length of drift region (source side)	(199)
RDRBBD	-		1		high field mobility in drift region (drain side)	(190)
RDRBBS	-		1		high field mobility in drift region (source side)	(190)
RDRMUED	-		1e3		mobility in drift region (drain side)	(186)
RDRMUES	-		1e3		mobility in drift region (source side)	(186)
RDRVMAXD	-		3e7		saturation velocity in drift region (drain side)	(188)
RDRVMAXS	-		3e7		saturation velocity in drift region (source side)	(188)
RDRVTMP	-		0		temperature dependence of resistance	(188)
RDRMUETMP	-		0		temperature dependence of resistance	(186)
RDRBBTMP	-		0		temperature dependence of resistance	(190)
RSH	$\Omega/\square$	$0 \leq$	0	$\leq 10 m$	source/drain sheet resistance	(191)
NOVERS	cm^{-3}		1e19		impurity concentration in overlap region (source side)	(198)
RDRDJUNC	m		1 u		junction depth at channel/drift region (drain side)	(196)
RDRVMAXL	-		0		saturation velocity L_{gate} dependence	(195)
RDRVMAXLP	-		1		saturation velocity L_{gate} dependence	(195)
RDRVMAXW	-		0		saturation velocity W_{gate} dependence	(195)
RDRVMAXWP	-		1		saturation velocity W_{gate} dependence	(195)
RDRMUEL	-		0		mobility in drift region L_{gate} dependence	(194)
RDRMUELP	-		1		mobility in drift region L_{gate} dependence	(194)

Parameter	Unit	Min	Default	Max	Description	See
Capacitance						
XQY	m	$0 \leq$	0	≤ 50 n	distance from junction to max field point	(204)
XQY1	$F \cdot \mu m^{XQY2-1}$	$0 \leq$	0		V_{es} -dependence of Q_y	(204)
XQY2	-	$0 \leq$	2		L_{gate} -dependence of Q_y	(204)
LOVER	m	$0 <$	30 n		overlap length	(??)
NOVER	cm^{-3}		1E19		impurity concentration in overlap region	(??)
VFBOVER	V		0		flat-band voltage in overlap region	(??)
CGDO	$F \cdot m^{-1}$	$0 \leq$	none	$\leq 10^2 nm \cdot C_{FOX}$	gate-to-drain overlap cap (not used unless specified)	p. 36
CGSO	$F \cdot m^{-1}$	$0 \leq$	none	$\leq 10^2 nm \cdot C_{FOX}$	gate-to-source overlap cap (not used unless specified)	p. 36
CGBO	$F \cdot m^{-1}$	$0 \leq$	none		gate-to-bulk overlap cap (not used unless specified)	(210)
Substrate current						
VFBSUB	V		-1		flatband voltage for I_{sub} calculation	(222)
VFBSUBL	$\mu m^{VFBSUBLP}$		0		L_{gate} -dependence of VFBSUB	(222)
VFBSUBLP	-		1		L_{gate} -dependence of VFBSUB	(222)
SUB1	V^{-1}		10		substrate current coefficient of magnitude	(224)
SUB1L	m		$2.5 \cdot 10^{-3}$		L_{gate} -dependence of SUB1	(224)
SUB1LP	-		1		L_{gate} -dependence SUB1	(224)
SUB2	V		20		substrate current coefficient of exponential term	(225)
SUB2L	m	$0 \leq$	$2 \cdot 10^{-6}$	≤ 1	L_{gate} -dependence of SUB2	(225)
SUBDLT	-		$2 \cdot 10^{-3}$		smoothing parameter	
SVDS	-		0.8		substrate current dependence on V_{ds}	(226)
SLG	m		$3 \cdot 10^{-8}$		substrate current dependence on L_{gate}	(226)
SVGS	-		0.8		substrate current dependence on V_{gs}	(228)
SVGSL	m^{SVGSLP}		0		L_{gate} -dependence of SVGS	(228)
SVGSLP	-		1		L_{gate} -dependence of SVGS	(228)
SVGSW	m^{SVGSWP}		0		W_{gate} -dependence of SVGS	(226)
SVGSWP	-		1		W_{gate} -dependence of SVGS	(226)
SVBS	-		0.5		substrate current dependence on V_{es}	(229)
SVBSL	m^{SVBSLP}		0		L_{gate} -dependence of SVBS	(229)
SVBSLP	-		1		L_{gate} -dependence of SVBS	(229)
IBPC1	$V \cdot A^{-1}$	$0 \leq$	0	$\leq 1 \cdot 10^{12}$	impact-ionization-induced bulk potential change	(231)
IBPC2	V^{-1}	$0 \leq$	0	$\leq 1 \cdot 10^{12}$	impact-ionization-induced bulk potential change	(231)
Gate leakage current						
GLEAK1	$V^{-3/2} \cdot s^{-1}$		$1 \cdot 10^4$		gate-to-channel current coefficient	(16.2)
GLEAK2	$V^{-1/2} \cdot m^{-1}$		20 MEG		gate-to-channel current coefficient	(16.2)
GLEAK3	-		$3 \cdot 10^{-1}$		gate-to-channel current coefficient	(233)
GLEAK4	m^{-1}		0		gate-to-channel current coefficient	(233)
GLEAK5	$V \cdot m^{-1}$		$7.5 \cdot 10^3$		G-t-C short channel correction	(233)
GLEAK6	V		$2.5 \cdot 10^{-1}$		G-t-C V_{ds} -dependence correction	(16.2)
GLEAK7	m^2		$1 \cdot 10^{-6}$		G-t-C L and W dependence correction	(16.2)
GLEAK8	m^2		1.0		gate-to-channel current coefficient	
GLEAK9	m^2		$5.0 \cdot 10^{-1}$		gate-to-channel current coefficient	
GLEAK10	m^2		0.0		gate-to-channel current coefficient	
GLKSD1	$A \cdot m \cdot V^{-2}$		1f		G-t-S/D current coefficient	(245)
GLKSD2	$V^{-1} \cdot m^{-1}$		5 MEG		G-t-S/D current coefficient	(245)
GLKSD3	m^{-1}		-5 MEG		G-t-S/D current coefficient	(245)
GLKSD4			0.0		G-t-S/D current coefficient	
GLKSD5			1.0		G-t-S/D current coefficient	
GLKB1	$A \cdot V^{-2}$		$5 \cdot 10^{-16}$		G-t-S/D current coefficient	(239)
GLKB2	$m \cdot V^{-1}$		1		G-t-S/D current coefficient	(239)
GLKB3	V		0		G-t-S/D current coefficient	(240)
GLKB4			1.0		G-t-S/D current coefficient	
GLKB5			1.0		G-t-S/D current coefficient	
GLKB6			1.0		G-t-S/D current coefficient	
GLKB7			$10 \cdot 10^{-12}$		G-t-S/D current coefficient	
GLKB8			$15 \cdot 10^{-12}$		G-t-S/D current coefficient	
GLKB21			$5 \cdot 10^{-16}$		G-t-S/D current coefficient	
GLKB22			1.0		G-t-S/D current coefficient	
GLKB23			0.0		G-t-S/D current coefficient	
GLKB24			1.0		G-t-S/D current coefficient	
GLKB25			1.0		G-t-S/D current coefficient	
GLKB26			1.0		G-t-S/D current coefficient	
GLKB27			$10 \cdot 10^{-12}$		G-t-S/D current coefficient	
GLKB28			$15 \cdot 10^{-12}$		G-t-S/D current coefficient	
FVBS	-	$0 \leq$	0		V_{es} dependence of Fowler-Nordheim current	(256)
EVB1	$V^{-2} s^{-1}$	$0 \leq$	0		electron tunneling from valence band	(255)
EVB2	$V \cdot m^{-1}$	$0 \leq$	0		electron tunneling from valence band	(255)
EVB3	-	$0 \leq$	0		electron tunneling from valence band	(256)

Parameter	Unit	Min	Default	Max	Description	See
Gate-induced drain/source leakage						
GIDL1	$\text{A} \cdot \text{V}^{-3/2} \text{C}^{-1} \text{m}^{1/2}$		$5 \cdot 10^{-6}$		magnitude of GIDL	(251)
GIDL2	$\text{V}^{-2} \text{m}^{-1} \text{F}^{-3/2}$		$1 \cdot 10^6$		field-dependence of GIDL	(251)
GIDL3	-		0.3		V_{ds} -dependence of GIDL	(253)
GIDL4*	V		0		threshold for V_{ds} dependence	(253)
GIDL5*	-		0.2		high-field correction	(254)
GIDLBPL1	m		$1.0 \cdot 10^{-6}$		length for GIDL/GISL parasitic bipolar effect	
GIDLBPLT			0.0		temperature effect for GIDLBPL1	
TFOXGIDL	m		$3.5 \cdot 10^{-9}$		TFOX for GIDL	
Floating-body effect						
QHE1	-	$0 <$	1.5		FBE parameter	(259)
QHE2	V	$0 <$	0.55		FBE parameter	(259)
HIST1	V	$0 <$	$1 \cdot 10^{-8}$		history-effect parameter	(266)
HIST2	A	$0 <$	$1 \cdot 10^{-20}$		history-effect parameter	(266)
$1/f$ noise						
NFALP	$\text{cm} \cdot \text{s}$		$1 \cdot 10^{-16}$		contribution of mobility fluctuation	(268)
NFTRP	V^{-1}		$1 \cdot 10^{10}$		ratio of trap density to attenuation coefficient	(268)
CIT*	$\text{F} \cdot \text{cm}^{-2}$		0		cap caused by the interface trapped carriers	(269)
FALPH	s m^3		1.0		power of f describing deviation of $1/f$	
Non-quasi-static model						
DLY1	s		$1 \cdot 10^{-10}$		coefficient for delay due to diffusion of carriers	(280)
DLY2	-		0.7		coefficient for delay due to conduction of carriers	(281)
DLY3	Ω		$8 \cdot 10^{-7}$		coefficient for RC delay of bulk carriers	(283)
Self-heating						
RTH0	KcmW^{-1}		0.1		thermal resistance	p. 48
CTH0	$\text{Ws}^{-(Kcm)}$		$1 \cdot 10^{-7}$		thermal capacitance	p. 48
Conservation of symmetry at $V_{\text{ds}} = 0$ for short-channel MOSFETs						
VZADD0	V		10m		symmetry conservation coefficient	
PZADD0	V		5m		symmetry conservation coefficient	

26 List of instance parameters, default values, and ranges

The maximum and minimum limits of the instance parameters are recommended values.

Parameter	Unit	Min	Default	Max	Description	See
L	m	$0 <$	$5 \cdot 10^{-6}$		gate length (L_{gate})	(1)
W	m	$0 <$	$5 \cdot 10^{-6}$		gate width (W_{gate})	(2)
XGL	m	$-L <$	0	$< L$	offset of gate length	(203)
XGW	m	$0 \leq$	0		distance from gate contact to channel edge	(203)
M	-	$0 <$	1		multiplication factor	p. 52
NF	-	$0 <$	1		number of gate fingers	(4)
NGCON	-	$0 <$	1		number of gate contacts	(203)
SA	m	$0 \leq$	0		length of diffusion between gate and STI	p. 29
SB	m	$0 \leq$	0		length of diffusion between gate and STI	p. 29
SD	m	$0 \leq$	0		length of diffusion between gate and gate	p. 29
SCA	[—]		0		layout characterization factor for WPE	(159)
SCB	[—]		0		layout characterization factor for WPE	(159)
SCC	[—]		0		layout characterization factor for WPE	(159)
TEMP	°C		27		device temperature	(174)
DTEMP	°C		0		device temperature change	(174)
NRD	-		1		number of drain squares	(191)
NRS	-		1		number of source squares	(197)
LDRIFT	m		1 u		length of drift region (drain side)	(193)
LDRIFTS	m		1 u		length of drift region (source side)	(199)

27 Overview of the Parameter-Extraction Procedure

In HiSIM_SOTB, device characteristics are strongly dependent on basic device parameter values, such as the impurity concentration and the oxide thickness. Therefore, the parameter-value extraction has to be repeated with measured characteristics of different devices in a specific sequence until extracted parameter values reproduce all device characteristics consistently and reliably. To achieve reliable results, it is recommended to start with initial parameter values according to the recommendations listed in the table below. Since some of the model parameters such as T_{FOX} are difficult to extract, they are expected to be determined directly by dedicated measurements. Threshold voltage measurements allow to derive a rough extraction for the model parameters referred to as “basic device parameters”.

Determined by dedicated measurements (not changed during extraction procedure)	Default values listed in the Section 25 are used initially for the groups of parameters listed below
TFOX TBOX TSOI	basic device parameters (not listed on left side) velocity quantum effect substrate current gate leakage GIDL mobility noise subthreshold swing non-quasi-static model overlap capacitances

The sequence of device selection for the parameter extraction is recommended in 4 steps

1. Long-channel devices
2. Short-channel devices
3. Long-narrow devices
4. Short-narrow devices
5. Temperature calibration

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